Chapter VI
Global Class Field Theory

§ 1. Idèles and Idèle Classes

The rôle held in local class field theory by the multiplicative group of the base field is taken in global class field theory by the idèle class group. The notion of idèle is a modification of the notion of ideal. It was introduced by the French mathematician *Claude Chevalley* (1909–1984) with a view to providing a suitable basis for the important local-to-global principle, i.e., for the principle which reduces problems concerning a number field K to analogous problems for the various completions $K_\mathfrak{p}$. *Chevalley* used the term "ideal element", which was abbreviated as id. el.

An **adèle** of K – this curious expression, which has the stress on the second syllable, is derived from the original term "additive idèle" – is a family

$$\alpha = (\alpha_\mathfrak{p})$$

of elements $\alpha_\mathfrak{p} \in K_\mathfrak{p}$ where $\mathfrak{p}$ runs through all primes of K, and $\alpha_\mathfrak{p}$ is integral in $K_\mathfrak{p}$ for almost all $\mathfrak{p}$. The adèles form a ring, which is denoted by

$$\mathbb{A}_K = \prod_\mathfrak{p} K_\mathfrak{p}.$$

Addition and multiplication are defined componentwise. This kind of product is called the "restricted product" of the $K_\mathfrak{p}$ with respect to the subrings $\mathcal{O}_\mathfrak{p} \subseteq K_\mathfrak{p}$.

The **idèle group** of K is defined to be the unit group

$$I_K = \mathbb{A}_K^*.$$

Thus an **idèle** is a family

$$\alpha = (\alpha_\mathfrak{p})$$

of elements $\alpha_\mathfrak{p} \in K_\mathfrak{p}^*$ where $\alpha_\mathfrak{p}$ is a unit in the ring $\mathcal{O}_\mathfrak{p}$ of integers of $K_\mathfrak{p}$, for almost all $\mathfrak{p}$. In analogy with $\mathbb{A}_K$, we write the idèle group as the restricted product

$$I_K = \prod_\mathfrak{p} K_\mathfrak{p}^*$$

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with respect to the unit groups $\mathcal{O}_\mathfrak{p}^*$. For every finite set of primes S, I_K contains the subgroup

$$I_K^S = \prod_{\mathfrak{p}\in S} K_\mathfrak{p}^* \times \prod_{\mathfrak{p}\notin S} U_\mathfrak{p}$$

of **S-idèles**, where $U_\mathfrak{p} = K_\mathfrak{p}^*$ for $\mathfrak{p}$ infinite complex, and $U_\mathfrak{p} = \mathbb{R}_+^*$ for $\mathfrak{p}$ infinite real. One clearly has

$$I_K = \bigcup_S I_K^S ,$$

if S varies over all finite sets of primes of K.

The inclusions $K \subseteq K_\mathfrak{p}$ allow us to define the diagonal embedding

$$K^* \longrightarrow I_K ,$$

which associates to $a \in K^*$ the idèle $\alpha \in I_K$ whose $\mathfrak{p}$-th component is the element a in $K_\mathfrak{p}$. We thus view K^* as a subgroup of I_K and we call the elements of K^* in I_K **principal idèles**. The intersection

$$K^S = K^* \cap I_K^S$$

consists of the numbers $a \in K^*$ which are units at all primes $\mathfrak{p} \notin S$, $\mathfrak{p} \nmid \infty$, and which are positive in $K_\mathfrak{p} = \mathbb{R}$ for all real infinite places $\mathfrak{p} \notin S$. They are called **S-units**. In particular, for the set S_∞ of infinite places, K^{S_∞} is the unit group $\mathcal{O}_K^*$ of $\mathcal{O}_K$. We get the following generalization of Dirichlet's unit theorem.

(1.1) Proposition. *If S contains all infinite places, then the homomorphism*

$$\lambda : K^S \longrightarrow \prod_{\mathfrak{p}\in S} \mathbb{R}, \quad \lambda(a) = \big(\log |a|_\mathfrak{p}\big)_{\mathfrak{p}\in S},$$

has kernel $\mu(K)$, and its image is a complete lattice in the $(s-1)$-dimensional trace-zero space $H = \big\{ (x_\mathfrak{p}) \in \prod_{\mathfrak{p}\in S} \mathbb{R} \mid \sum_{\mathfrak{p}\in S} x_\mathfrak{p} = 0 \big\}$, $s = \#S$.

Proof: For the set $S_\infty = \{\mathfrak{p} \mid \infty\}$, this is the claim of chap. I, (7.1) and (7.3). Let $S_\mathrm{f} = S \smallsetminus S_\infty$, and let $J(S_\mathrm{f})$ be the subgroup of J_K generated by the prime ideals $\mathfrak{p} \in S_\mathrm{f}$. Associating to every $a \in K^S$ the principal ideal $ia = (a) \in J(S_\mathrm{f})$, we obtain the commutative diagram

$$\begin{tikzcd}
1 \arrow[r] & \mathcal{O}_K^* \arrow[r] \arrow[d, "\lambda'"] & K^S \arrow[r, "i"] \arrow[d, "\lambda"] & J(S_\mathrm{f}) \arrow[d, "\lambda''"] \\
0 \arrow[r] & \prod\limits_{\mathfrak{p}\in S_\infty} \mathbb{R} \arrow[r] & \prod\limits_{\mathfrak{p}\in S} \mathbb{R} \arrow[r, "i"] & \prod\limits_{\mathfrak{p}\in S_\mathrm{f}} \mathbb{R}
\end{tikzcd}$$

with exact rows. The map λ'' on the right is given by

$$\lambda''\Big(\prod_{\mathfrak{p}\in S_f} \mathfrak{p}^{\nu_\mathfrak{p}}\Big) = -\prod_{\mathfrak{p}\in S_f} \nu_\mathfrak{p} \log \mathfrak{N}(\mathfrak{p})$$

(observe that $|a|_\mathfrak{p} = \mathfrak{N}(\mathfrak{p})^{-\nu_\mathfrak{p}(a)}$), and maps $J(S_f)$ isomorphically onto the complete lattice spanned by the vectors

$$e_\mathfrak{p} = \big(0, \dots, 0, \log \mathfrak{N}(\mathfrak{p}), 0, \dots, 0\big),$$

for $\mathfrak{p} \in S_f$. It follows that $\ker(\lambda) = \ker(\lambda') = \mu(K)$, and we obtain the exact sequence

$$(*) \qquad 0 \longrightarrow \operatorname{im}(\lambda') \longrightarrow \operatorname{im}(\lambda) \xrightarrow{\ i\ } \operatorname{im}(\lambda''),$$

where the groups on the left and on the right are lattices. This implies that the group in the middle is also a lattice. For if $x \in \operatorname{im}(\lambda)$, and U is a neighbourhood of $i(x)$ which contains no other point of $\operatorname{im}(\lambda'')$, then $i^{-1}(U)$ contains the coset $x + \operatorname{im}(\lambda')$, and no other. It is discrete since $\operatorname{im}(\lambda')$ is discrete.

For every $\mathfrak{p} \in S_f$, if h is the class number of K, then $\mathfrak{p}^h$ belongs to $i(K^S)$, i.e.,

$$J(S_f)^h \subseteq i(K^S) \subseteq J(S_f).$$

The groups on the left and on the right have rank $\#S_f$, hence so does $i(K^S)$. In the sequence $(*)$, the image of i therefore has rank $\#S_f$, and the kernel has rank $\#S_\infty - 1$. Hence $\operatorname{im}(\lambda)$ is a lattice of rank $\#S_\infty - 1 + \#S_f = \#S - 1$. It lies in the $(\#S-1)$-dimensional trace-zero space H, since $\prod_{\mathfrak{p}\in S} |a|_\mathfrak{p} = \prod_\mathfrak{p} |a|_\mathfrak{p} = 1$ for $a \in K^S$. □

(1.2) Definition. *The elements of the subgroup K^* of I_K are called* **principal idèles** *and the quotient group*

$$C_K = I_K/K^*$$

is called the **idèle class group** *of K.*

The relation between the ideal class group Cl_K and the idèle class group C_K is as follows. There is a surjective homomorphism

$$(\ ): I_K \longrightarrow J_K, \quad \alpha \longmapsto (\alpha) = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\nu_\mathfrak{p}(\alpha_\mathfrak{p})},$$

from the idèle group I_K to the ideal group J_K. Its kernel is

$$I_K^{S_\infty} = \prod_{\mathfrak{p}|\infty} K_\mathfrak{p}^* \times \prod_{\mathfrak{p}\nmid\infty} U_\mathfrak{p}.$$

It induces a surjective homomorphism

$$C_K \longrightarrow Cl_K$$

with kernel $I_K^{S_\infty}K^*/K^*$. We may also consider the surjective homomorphism

$$I_K \longrightarrow J(\bar{\mathcal{O}}), \quad \alpha \longmapsto \prod_{\mathfrak{p}} \mathfrak{p}^{v_\mathfrak{p}(\alpha_\mathfrak{p})},$$

onto the *replete* ideal group $J(\bar{\mathcal{O}})$. Its kernel is

$$I_K^0 = \big\{ (\alpha_\mathfrak{p}) \in I_K \bigm| |\alpha_\mathfrak{p}|_\mathfrak{p} = 1 \text{ for all } \mathfrak{p} \big\}$$

(see chap. III, § 1). It takes principal idèles to replete principal ideals and induces a surjective homomorphism

$$C_K \longrightarrow Pic(\bar{\mathcal{O}})$$

onto the replete ideal class group, with kernel $I_K^0 K^*/K^*$. We therefore have the

(1.3) Proposition. $Cl_K \cong I_K/I_K^{S_\infty}K^*$, *and* $Pic(\bar{\mathcal{O}}) \cong I_K/I_K^0 K^*$.

In contrast to the ideal class group, the idèle class group is not finite. But the finiteness of the former is reflected in terms of the latter as follows.

(1.4) Proposition. $I_K = I_K^S K^*$, i.e., $C_K = I_K^S K^*/K^*$, *if* S *is a sufficiently big finite set of places of* K.

Proof: Let $\mathfrak{a}_1, \dots, \mathfrak{a}_h$ be ideals representing the h classes of J_K/P_K. They are composed of a finite number of prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$. Now if S is any finite set of places containing these primes and the places at infinity, then one has $I_K = I_K^S K^*$.

In order to see this, we use the isomorphism $I_K/I_K^{S_\infty} \cong J_K$. If $\alpha \in I_K$, then the corresponding ideal $(\alpha) = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{v_\mathfrak{p}(\alpha_\mathfrak{p})}$ belongs to some class $\mathfrak{a}_i P_K$, i.e., $(\alpha) = \mathfrak{a}_i(a)$ for some principal ideal (a). The idèle $\alpha' = \alpha a^{-1}$ is mapped by $I_K \to J_K$ to the ideal $\mathfrak{a}_i = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{v_\mathfrak{p}(\alpha'_\mathfrak{p})}$. Since the prime ideals occurring in $\mathfrak{a}_i$ lie in S, we have $v_\mathfrak{p}(\alpha'_\mathfrak{p}) = 0$, i.e., $\alpha'_\mathfrak{p} \in U_\mathfrak{p}$ for all $\mathfrak{p} \notin S$. Hence $\alpha' = \alpha a^{-1} \in I_K^S$, and thus $\alpha \in I_K^S K^*$. $\square$

The idèle group comes equipped with a canonical topology. A basic system of neighbourhoods of $1 \in I_K$ is given by the sets

$$\prod_{\mathfrak{p} \in S} W_{\mathfrak{p}} \times \prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}} \subseteq I_K ,$$

where S runs through the finite sets of places of K which contain all $\mathfrak{p}|\infty$, and $W_{\mathfrak{p}} \subseteq K_{\mathfrak{p}}^*$ is a basic system of neighbourhoods of $1 \in K_{\mathfrak{p}}^*$. The groups $U_{\mathfrak{p}}$ are compact for $\mathfrak{p} \notin S$. Therefore the same is true of the group $\prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}}$. If the $W_{\mathfrak{p}}$, for $\mathfrak{p}|\infty$, are bounded, then $\prod_{\mathfrak{p} \in S} W_{\mathfrak{p}} \times \prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}}$ is a neighbourhood of 1 in I_K whose closure is compact. Therefore I_K is a **locally compact topological group**.

(1.5) Proposition. *K^* is a discrete, and therefore closed, subgroup of I_K.*

Proof: It is enough to show that $1 \in I_K$ has a neighbourhood which contains no other principal idèle besides 1.

$$\mathfrak{U} = \left\{ \alpha \in I_K \;\middle|\; |\alpha_{\mathfrak{p}}|_{\mathfrak{p}} = 1 \text{ for } \mathfrak{p} \nmid \infty,\ |\alpha_{\mathfrak{p}} - 1|_{\mathfrak{p}} < 1 \text{ for } \mathfrak{p}|\infty \right\},$$

is such a neighbourhood. For if we had a principal idèle $x \in \mathfrak{U}$ different from 1, then we get the contradiction

$$\begin{aligned} 1 = \prod_{\mathfrak{p}} |x-1|_{\mathfrak{p}} &= \prod_{\mathfrak{p} \nmid \infty} |x-1|_{\mathfrak{p}} \cdot \prod_{\mathfrak{p}|\infty} |x-1|_{\mathfrak{p}} \\ &< \prod_{\mathfrak{p} \nmid \infty} |x-1|_{\mathfrak{p}} \le \prod_{\mathfrak{p} \nmid \infty} \max\{ |x|_{\mathfrak{p}}, 1\} = 1 . \end{aligned}$$

That the subgroup is closed follows for a completely general reason: since $(x,y) \mapsto xy^{-1}$ is continuous, there is a neighbourhood V of 1 such that $VV^{-1} \subseteq \mathfrak{U}$. For every $y \in I_K$, the neighbourhood yV then contains at most one $x \in K^*$. Indeed, from $x_1 = yv_1$, $x_2 = yv_2 \in K^*$, with $x_1 \neq x_2$, one deduces $x_1 x_2^{-1} = v_1 v_2^{-1} \in \mathfrak{U}$, a contradiction. □

As K^* is closed in I_K, the fact that I_K is a locally compact Hausdorff topological group carries over to the idèle class group $C_K = I_K/K^*$. For any idèle $\alpha = (\alpha_{\mathfrak{p}}) \in I_K$, its class in C_K will be denoted by $[\alpha]$. We define the **absolute norm** of α to be the real number

$$\mathfrak{N}(\alpha) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{v_{\mathfrak{p}}(\alpha_{\mathfrak{p}})} = \prod_{\mathfrak{p}} |\alpha_{\mathfrak{p}}|_{\mathfrak{p}}^{-1} .$$

If $x \in K^*$ is a principal idèle, then we find by chap. III, (1.3), that $\mathfrak{N}(x) = \prod_{\mathfrak{p}} |x|_{\mathfrak{p}}^{-1} = 1$. We thus have a continuous homomorphism

$$\mathfrak{N} : C_K \longrightarrow \mathbb{R}_+^* .$$

It is related to the absolute norm on the replete Picard group $Pic(\bar{o})$ via the commutative diagram

$$\begin{array}{ccc} C_K & \xrightarrow{\mathfrak{N}} & \mathbb{R}_+^* \\ \downarrow & & \| \\ Pic(\bar{o}) & \xrightarrow{\mathfrak{N}} & \mathbb{R}_+^* . \end{array}$$

Here the arrow

$$C_K \longrightarrow Pic(\bar{o})$$

is induced by the continuous surjective homomorphism

$$I_K \longrightarrow J(\bar{o}), \quad (\alpha_\mathfrak{p}) \longmapsto \prod_\mathfrak{p} \mathfrak{p}^{v_\mathfrak{p}(\alpha_\mathfrak{p})},$$

with kernel

$$I_K^0 = \big\{ (\alpha_\mathfrak{p}) \in I_K \bigm| |\alpha_\mathfrak{p}|_\mathfrak{p} = 1 \text{ for all } \mathfrak{p} \big\}.$$

As to the kernel C_K^0 of $\mathfrak{N} : C_K \to \mathbb{R}_+$, we obtain, in analogy with chap. III, (1.14), the following important theorem. It reflects the finiteness of the unit rank of K as well as the finiteness of the class number.

(1.6) Theorem. *The group* $C_K^0 = \{[\alpha] \in C_K \mid \mathfrak{N}([\alpha]) = 1\}$ *is compact.*

Proof: The claim concerning the commutative exact diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & C_K^0 & \longrightarrow & C_K & \longrightarrow & \mathbb{R}_+^* & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \to & Pic(\bar{o})^0 & \longrightarrow & Pic(\bar{o}) & \longrightarrow & \mathbb{R}_+^* & \longrightarrow & 1 \end{array}$$

will be reduced to the compactness of the group $Pic(\bar{o})^0$, which was proved in chap. III, (1.14). The kernel of the vertical arrow in the middle is the group $I_K^0 K^*/K^* = I_K^0 / I_K^0 \cap K^*$, where we have $I_K^0 = \prod_\mathfrak{p} I_\mathfrak{p}^0$, $I_\mathfrak{p}^0 = \{\alpha_\mathfrak{p} \in K_\mathfrak{p} \mid |\alpha_\mathfrak{p}|_\mathfrak{p} = 1\}$, and $I_K^0 \cap K^* = \mu(K)$ by chap. III, (1.9). This kernel is clearly compact. We obtain an exact sequence

$$1 \longrightarrow I_K^0 K^*/K^* \longrightarrow C_K^0 \longrightarrow Pic(\bar{o})^0 \longrightarrow 1$$

of continuous homomorphisms. Since $Pic(\bar{o})^0$ is compact, and the same is true for the fibres of the mapping $C_K^0 \to Pic(\bar{o})^0$ (they are cosets, all homeomorphic to $I_K^0 K^*/K^*$), hence so is C_K^0. □

The idèle class group C_K plays a similar rôle for the algebraic number field K as the multiplicative group $K_\mathfrak{p}^*$ does for a $\mathfrak{p}$-adic number field $K_\mathfrak{p}$. It comes equipped with a collection of canonical subgroups which are to be viewed as analogues of the higher unit groups $U_\mathfrak{p}^{(n)} = 1 + \mathfrak{p}^n$ of a $\mathfrak{p}$-adic number field $K_\mathfrak{p}$. Instead of $\mathfrak{p}^n$, we take any integral ideal $\mathfrak{m} = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{n_\mathfrak{p}}$. We may also write it as a replete ideal

$$\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$$

with $n_\mathfrak{p} = 0$ for $\mathfrak{p}|\infty$, and we treat it in what follows as a **module** of K. For every place $\mathfrak{p}$ of K we put $U_\mathfrak{p}^{(0)} = U_\mathfrak{p}$, and

$$U_\mathfrak{p}^{(n_\mathfrak{p})} := \begin{cases} 1 + \mathfrak{p}^{n_\mathfrak{p}}, & \text{if } \mathfrak{p} \nmid \infty, \\ \mathbb{R}_+^* \subset K_\mathfrak{p}^*, & \text{if } \mathfrak{p} \text{ is real}, \\ \mathbb{C}^* = K_\mathfrak{p}^*, & \text{if } \mathfrak{p} \text{ is complex}, \end{cases}$$

for $n_\mathfrak{p} > 0$. Given $\alpha_\mathfrak{p} \in K_\mathfrak{p}^*$ we write

$$\alpha_\mathfrak{p} \equiv 1 \bmod \mathfrak{p}^{n_\mathfrak{p}} \iff \alpha_\mathfrak{p} \in U_\mathfrak{p}^{(n_\mathfrak{p})}.$$

For a finite prime $\mathfrak{p}$ and $n_\mathfrak{p} > 0$ this means the usual congruence; for a real place, it symbolizes positivity, and for a complex place it is the empty condition.

(1.7) Definition. *The group*

$$C_K^\mathfrak{m} = I_K^\mathfrak{m} K^*/K^*,$$

formed from the idèle group

$$I_K^\mathfrak{m} = \prod_{\mathfrak{p}} U_\mathfrak{p}^{(n_\mathfrak{p})},$$

is called the **congruence subgroup** mod $\mathfrak{m}$, *and the quotient group* $C_K/C_K^\mathfrak{m}$ *is called the* **ray class group** mod $\mathfrak{m}$.

Remark: This definition of the ray class group does correspond to the classical one, as given (in the ideal-theoretic version) for instance in Hasse's "Zahlbericht" [53]. It differs from those found in modern textbooks, and also from that given in [107] by the author: in the present book, the components $\alpha_\mathfrak{p}$ of idèles α in $I_K^\mathfrak{m}$ are always positive at all real places $\mathfrak{p}$, so we have here fewer congruence subgroups than in the other texts. This choice does not only simplify matters. Most of all, it was made substantially because of the choice of the canonical metric $\langle\ ,\ \rangle$ on the Minkowski space $K_{\mathbb{R}}$ (see chap. I, § 5). In fact, we saw in chap. III, § 3, that this choice forces the extension $\mathbb{C}|\mathbb{R}$ to be unramified. We will explain in § 6 below how to interpret this situation, and how to reconcile it with the definition of ray classes in other texts.

The significance of the congruence subgroups lies in that they provide an overview over all closed subgroups of finite index in C_K. More precisely, we have the

(1.8) Proposition. *The closed subgroups of finite index of C_K are precisely those subgroups that contain a congruence subgroup $C_K^{\mathfrak{m}}$.*

Proof: $C_K^{\mathfrak{m}}$ is open in C_K because $I_K^{\mathfrak{m}} = \prod_{\mathfrak{p}} U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$ is open in I_K. $I_K^{\mathfrak{m}}$ is contained in the group $I_K^{S_\infty} = \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}^* \times \prod_{\mathfrak{p}\nmid\infty} U_{\mathfrak{p}}$, and since $(C_K : I_K^{S_\infty}K^*/K^*) = \# Cl_K = h < \infty$, the index

$$\begin{aligned}(C_K : C_K^{\mathfrak{m}}) &= h(I_K^{S_\infty}K^* : I_K^{\mathfrak{m}}K^*) \le h(I_K^{S_\infty} : I_K^{\mathfrak{m}})\\ &= h \prod_{\mathfrak{p}\nmid\infty} (U_{\mathfrak{p}} : U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}) \prod_{\mathfrak{p}|\infty} (K_{\mathfrak{p}}^* : U_{\mathfrak{p}}^{(n_{\mathfrak{p}})})\end{aligned}$$

is finite. Being the complement of the nontrivial open cosets, which are finite in number, $C_K^{\mathfrak{m}}$ is closed of finite index. Consequently, every group containing $C_K^{\mathfrak{m}}$ is also closed of finite index, for it is the union of finitely many cosets of $C_K^{\mathfrak{m}}$.

Conversely, let $\mathcal{N}$ be an arbitrary closed subgroup of finite index. Then $\mathcal{N}$ is also open, being the complement of a finite number of closed cosets. Thus the preimage J of $\mathcal{N}$ in I_K is also open, and it thus contains a subset of the form

$$W = \prod_{\mathfrak{p}\in S} W_{\mathfrak{p}} \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}},$$

where S is a finite set of places of K containing the infinite ones, and $W_{\mathfrak{p}}$ is an open neighbourhood of $1 \in K_{\mathfrak{p}}^*$. If $\mathfrak{p} \in S$ is finite, we are liable to choose $W_{\mathfrak{p}} = U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$, because the groups $U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} \subseteq K_{\mathfrak{p}}^*$ form a basic system of neighbourhoods of $1 \in K_{\mathfrak{p}}^*$. If $\mathfrak{p} \in S$ is real, we may choose $W_{\mathfrak{p}} \subseteq \mathbb{R}_+^*$. The open set $W_{\mathfrak{p}}$ will then generate the group $\mathbb{R}_+^*$, resp. $K_{\mathfrak{p}}^*$ in the case of a complex place $\mathfrak{p}$. The subgroup of J generated by W is therefore of the form $I_K^{\mathfrak{m}}$, so $\mathcal{N}$ contains the congruence subgroup $C_K^{\mathfrak{m}}$. □

The ray class groups can be given the following purely ideal-theoretic description. Let $J_K^{\mathfrak{m}}$ be the group of all fractional ideals relatively prime to $\mathfrak{m}$, and let $P_K^{\mathfrak{m}}$ be the group of all principal ideals $(a) \in P_K$ such that

$$a \equiv 1 \bmod \mathfrak{m} \quad \text{and} \quad a \text{ totally positive.}$$

The latter condition means that, for every real embedding $K \to \mathbb{R}$, a turns out to be positive. The congruence $a \equiv 1 \bmod \mathfrak{m}$ means that a is the quotient b/c of two *integers* relatively prime to $\mathfrak{m}$ such that $b \equiv c \bmod \mathfrak{m}$. This is tantamount to saying that $a \equiv 1 \bmod \mathfrak{p}^{n_\mathfrak{p}}$ in $K_\mathfrak{p}$, i.e., $a \in U_\mathfrak{p}^{(n_\mathfrak{p})}$ for all $\mathfrak{p}|\mathfrak{m} = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{n_\mathfrak{p}}$. We put

$$Cl_K^{\mathfrak{m}} = J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}.$$

We then have the

(1.9) Proposition. *The homomorphism*

$$(\ ): I_K \longrightarrow J_K, \quad \alpha \longmapsto (\alpha) = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{v_\mathfrak{p}(\alpha_\mathfrak{p})},$$

induces an isomorphism

$$C_K/C_K^{\mathfrak{m}} \cong Cl_K^{\mathfrak{m}}.$$

Proof: Let $\mathfrak{m} = \prod_\mathfrak{p} \mathfrak{p}^{n_\mathfrak{p}}$, and let

$$I_K^{(\mathfrak{m})} = \left\{\alpha \in I_K \mid \alpha_\mathfrak{p} \in U_\mathfrak{p}^{(n_\mathfrak{p})} \text{ for } \mathfrak{p}|\mathfrak{m}\infty\right\}.$$

Then $I_K = I_K^{(\mathfrak{m})}K^*$, because for every $\alpha \in I_K$, by the approximation theorem, there exists an $a \in K^*$ such that $\alpha_\mathfrak{p} a \equiv 1 \bmod \mathfrak{p}^{n_\mathfrak{p}}$ for $\mathfrak{p}|\mathfrak{m}$, and $\alpha_\mathfrak{p} a > 0$ for $\mathfrak{p}$ real. Thus $\beta = (\alpha_\mathfrak{p} a) \in I_K^{(\mathfrak{m})}$, so that $\alpha = \beta a^{-1} \in I_K^{(\mathfrak{m})}K^*$. The elements $a \in I_K^{(\mathfrak{m})} \cap K^*$ are precisely those generating principal ideals in $P_K^{\mathfrak{m}}$. Therefore the correspondence $\alpha \mapsto (\alpha) = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{v_\mathfrak{p}(\alpha_\mathfrak{p})}$ defines a surjective homomorphism

$$C_K = I_K^{(\mathfrak{m})}K^*/K^* = I_K^{(\mathfrak{m})}/I_K^{(\mathfrak{m})} \cap K^* \longrightarrow J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}.$$

Since $(\alpha) = 1$ for $\alpha \in I_K^{\mathfrak{m}}$, the group $C_K^{\mathfrak{m}} = I_K^{\mathfrak{m}}K^*/K^*$ is certainly contained in the kernel. Conversely, if the class $[\alpha]$ represented by $\alpha \in I_K^{(\mathfrak{m})}$ belongs to the kernel, then there is an $(a) \in P_K^{\mathfrak{m}}$, with $a \in I_K^{(\mathfrak{m})} \cap K^*$, such that $(\alpha) = (a)$. The components of the idèle $\beta = \alpha a^{-1}$ satisfy $\beta_\mathfrak{p} \in U_\mathfrak{p}$ for $\mathfrak{p} \nmid \mathfrak{m}\infty$, and $\beta_\mathfrak{p} \in U_\mathfrak{p}^{(n_\mathfrak{p})}$ for $\mathfrak{p}|\mathfrak{m}\infty$, in other words, $\beta \in I_K^{\mathfrak{m}}$, and hence $[\alpha] = [\beta] \in I_K^{\mathfrak{m}}K^*/K^* = C_K^{\mathfrak{m}}$. Therefore $C_K^{\mathfrak{m}}$ is the kernel of the above mapping, and the proposition is proved. □

The ray class groups in the ideal-theoretic version $Cl_K^{\mathfrak{m}} = J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ were introduced by *Heinrich Weber* (1842–1913) as a common generalization of ideal class groups on the one hand, and the groups $(\mathbb{Z}/m\mathbb{Z})^*$ on the other. These latter groups may be viewed as the ray class groups of the field $\mathbb{Q}$:

(1.10) Proposition. *For any module $\mathfrak{m} = (m)$ of the field $\mathbb{Q}$, one has*

$$C_{\mathbb{Q}}/C_{\mathbb{Q}}^{\mathfrak{m}} \cong Cl_{\mathbb{Q}}^{\mathfrak{m}} \cong (\mathbb{Z}/m\mathbb{Z})^* .$$

Proof: Every ideal $(a) \in J_{\mathbb{Q}}^{\mathfrak{m}}$ has two generators, a and $-a$. Mapping the *positive* generator onto the residue class $\bmod m$, we get a surjective homomorphism $J_{\mathbb{Q}}^{\mathfrak{m}} \to (\mathbb{Z}/m\mathbb{Z})^*$ whose kernel consists of all ideals (a) which have a positive generator $\equiv 1 \bmod m$. But these are precisely the ideals (a) such that $a \equiv 1 \bmod p^{n_p}$ for $p|m\infty$, i.e., the kernel of $P_{\mathbb{Q}}^{\mathfrak{m}}$. $\square$

The group $(\mathbb{Z}/m\mathbb{Z})^*$ is canonically isomorphic to the Galois group $G(\mathbb{Q}(\mu_m)|\mathbb{Q})$ of the m-th cyclotomic field $\mathbb{Q}(\mu_m)$. We therefore obtain a canonical isomorphism

$$G(\mathbb{Q}(\mu_m)|\mathbb{Q}) \cong C_{\mathbb{Q}}/C_{\mathbb{Q}}^{\mathfrak{m}} .$$

It is class field theory, which provides a far-reaching generalization of this important fact. For all modules $\mathfrak{m}$ of an arbitrary number field K, there will be Galois extensions $K^{\mathfrak{m}}|K$ generalizing the cyclotomic fields: the so-called **ray class fields**, which satisfy canonically

$$G(K^{\mathfrak{m}}|K) \cong C_K/C_K^{\mathfrak{m}}$$

(see § 6). The ray class group $\bmod 1$ is of particular interest here. It is related to the ideal class group Cl_K – which according to our definition here, is in general not a ray class group – as follows.

(1.11) Proposition. *There is an exact sequence*

$$1 \longrightarrow \mathcal{O}^*/\mathcal{O}_+^* \longrightarrow \prod_{\mathfrak{p} \text{ real}} \mathbb{R}^*/\mathbb{R}_+^* \longrightarrow Cl_K^1 \longrightarrow Cl_K \longrightarrow 1,$$

where $\mathcal{O}_+^*$ is the group of totally positive units of K.

Proof: One has $Cl_K^1 \cong C_K/C_K^1 = I_K/I_K^1K^*$ and, by (1.3), $Cl_K \cong I_K/I_K^{S_\infty}K^*$, where $I_K^1 = \prod_{\mathfrak{p}} U_{\mathfrak{p}}$ and $I_K^{S_\infty} = \prod_{\mathfrak{p}\nmid\infty} U_{\mathfrak{p}} \times \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}^*$. We therefore obtain an exact sequence

$$1 \longrightarrow I_K^{S_\infty}K^*/I_K^1K^* \longrightarrow C_K/C_K^1 \longrightarrow Cl_K \longrightarrow 1 .$$

For the group on the left we have the exact sequence

$$1 \longrightarrow I_K^{S_\infty} \cap K^*/I_K^1 \cap K^* \longrightarrow I_K^{S_\infty}/I_K^1 \longrightarrow I_K^{S_\infty}K^*/I_K^1K^* \longrightarrow 1.$$

But $I_K^{S_\infty} \cap K^* = \mathcal{O}^*$, $I_K^1 \cap K^* = \mathcal{O}_+^*$, and $I_K^{S_\infty}/I_K^1 = \prod_{\mathfrak{p}|\infty} K_\mathfrak{p}^*/U_\mathfrak{p} = \prod_{\mathfrak{p} \text{ real}} \mathbb{R}^*/\mathbb{R}_+^*$. □

Exercise 1. (i) $\mathbb{A}_\mathbb{Q} = (\widehat{\mathbb{Z}} \otimes_\mathbb{Z} \mathbb{Q}) \times \mathbb{R}$.

(ii) The quotient group $\mathbb{A}_\mathbb{Q}/\mathbb{Z}$ is compact and connected.

(iii) $\mathbb{A}_\mathbb{Q}/\mathbb{Z}$ is arbitrarily and uniquely divisible, i.e., the equation $nx = y$ has a unique solution, for every $n \in \mathbb{N}$ and $y \in \mathbb{A}_\mathbb{Q}/\mathbb{Z}$.

Exercise 2. Let K be a number field, $m = 2^\nu m'$ (m' odd), and let S be a finite set of primes. Let $a \in K^*$ and $a \in K_\mathfrak{p}^{*m}$, for all $\mathfrak{p} \notin S$. Show:

(i) If $K(\zeta_{2^\nu})|K$ is cyclic, where ζ_{2^ν} is a primitive 2^ν-th root of unity, then $a \in K^{*m}$.

(ii) Otherwise one has at least that $a \in K^{*m/2}$.

Hint: Use the following fact, proved in (3.8): if $L|K$ is a finite extension in which almost all prime ideals split completely, then $L = K$.

Exercise 3. Write $I_K^1 = I_\mathrm{f}^1 \times I_\infty^1$, with $I_\mathrm{f}^1 = \prod_{\mathfrak{p}\nmid\infty} U_\mathfrak{p}$, $I_\infty^1 = \prod_{\mathfrak{p}|\infty} U_\mathfrak{p}$. Show that taking integer powers of idèles $\alpha \in I_\mathrm{f}^1$ extends by continuity to exponentiation α^x with $x \in \widehat{\mathbb{Z}}$.

Exercise 4. Let $\varepsilon_1, \dots, \varepsilon_t \in \mathcal{O}_K^*$ be independent units. The images $\bar\varepsilon_1, \dots, \bar\varepsilon_t$ in I_f^1 are then independent units with respect to the exponentiation with elements of $\widehat{\mathbb{Z}}$, i.e., any relation

$$\bar\varepsilon_1^{x_1} \cdots \bar\varepsilon_t^{x_r} = 1, \quad x_i \in \widehat{\mathbb{Z}},$$

implies $x_i = 0$, $i = 1, \dots, t$.

Exercise 5. Let $\varepsilon \in \mathcal{O}_K^*$ be totally positive, i.e., $\varepsilon \in I_K^1$. Extend the exponentiation $\mathbb{Z} \to I_K^1$, $n \mapsto \varepsilon^n$, by continuity to an exponentiation $\widehat{\mathbb{Z}} \times \mathbb{R} \to I_K^1 = I_\mathrm{f}^1 \times I_\infty^1$, $\varepsilon \mapsto \varepsilon^\lambda$, in such a way that $\mathfrak{N}(\varepsilon^\lambda) = 1$.

Exercise 6. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_s$ be the complex primes of K. For $y \in \mathbb{R}$, let $\phi_k(y)$ be the idèle having component $e^{2\pi i y}$ at $\mathfrak{p}_k$, and components 1 at all other places. Let $\varepsilon_1, \dots, \varepsilon_t$ be a $\mathbb{Z}$-basis of the group of totally positive units of K.

(i) The idèles of the form

$$\alpha = \varepsilon_1^{\lambda_1} \cdots \varepsilon_t^{\lambda_t} \phi_1(y_1) \cdots \phi_s(y_s), \quad \lambda_i \in \widehat{\mathbb{Z}} \times \mathbb{R},\ y_i \in \mathbb{R},$$

form a group, and have absolute norm $\mathfrak{N}(\alpha) = 1$.

(ii) α is a principal ideal if and only if $\lambda_\iota \in \mathbb{Z} \subseteq \widehat{\mathbb{Z}} \times \mathbb{R}$ and $y_i \in \mathbb{Z} \subseteq \mathbb{R}$.

Exercise 7. Sending

$$(\lambda_1, \dots, \lambda_t, y_1, \dots, y_s) \mapsto \varepsilon_1^{\lambda_1} \cdots \varepsilon_t^{\lambda_t} \phi_1(y_1) \cdots \phi_s(y_s)$$

defines a continuous homomorphism

$$f : (\widehat{\mathbb{Z}} \times \mathbb{R})^t \times \mathbb{R}^s \longrightarrow C_K^0$$

into the group $C_K^0 = \{[\alpha] \in C_K \mid \mathfrak{N}([\alpha]) = 1\}$, with kernel $\mathbb{Z}^t \times \mathbb{Z}^s$.

Exercise 8. (i) The image D_K^0 of f is compact, connected and arbitrarily divisible.
(ii) f yields a topological isomorphism

$$f : ((\widehat{\mathbb{Z}} \times \mathbb{R})/\mathbb{Z})^t \times (\mathbb{R}/\mathbb{Z})^s \xrightarrow{\sim} D_K^0 .$$

Exercise 9. The group D_K^0 is the intersection of all closed subgroups of finite index in C_K^0, and it is the connected component of 1 in C_K^0.

Exercise 10. The connected component D_K of 1 in the idèle class group C_K is the direct product of t copies of the "solenoid" $(\widehat{\mathbb{Z}} \times \mathbb{R})/\mathbb{Z}$, s circles $\mathbb{R}/\mathbb{Z}$, and a real line.

Exercise 11. Every ideal class of the ray class group $Cl_K^{\mathfrak{m}}$ can be represented by an integral ideal which is prime to an arbitrary fixed ideal.

Exercise 12. Let $\mathcal{O} = \mathcal{O}_K$. Every class in $(\mathcal{O}/\mathfrak{m})^*$ can be represented by a totally positive number in $\mathcal{O}$ which is prime to an arbitrary fixed ideal.

Exercise 13. For every module $\mathfrak{m}$, one has an exact sequence

$$1 \to \mathcal{O}_+^*/\mathcal{O}_+^{\mathfrak{m}} \longrightarrow (\mathcal{O}/\mathfrak{m})^* \longrightarrow Cl_K^{\mathfrak{m}} \longrightarrow Cl_K^1 \to 1 ,$$

where $\mathcal{O}_+^*$, resp. $\mathcal{O}_+^{\mathfrak{m}}$, is the group of totally positive units of $\mathcal{O}$, resp. of totally positive units $\equiv 1 \bmod \mathfrak{m}$.

Exercise 14. Compute the kernels of $Cl_K^{\mathfrak{m}} \to Cl_K$ and $Cl_K^{\mathfrak{m}} \to Cl_K^{\mathfrak{m}'}$ for $\mathfrak{m}'|\mathfrak{m}$.

§ 2. Idèles in Field Extensions

We shall now study the behaviour of idèles and idèle classes when we pass from a field K to an extension L. So let $L|K$ be a finite extension of algebraic number fields. We embed the idèle group I_K of K into the idèle group I_L of L by sending an idèle $\alpha = (\alpha_{\mathfrak{p}}) \in I_K$ to the idèle $\alpha' = (\alpha'_{\mathfrak{P}}) \in I_L$ whose components $\alpha'_{\mathfrak{P}}$ are given by

$$\alpha'_{\mathfrak{P}} = \alpha_{\mathfrak{p}} \in K_{\mathfrak{p}}^* \subseteq L_{\mathfrak{P}}^* \quad \text{for} \quad \mathfrak{P}|\mathfrak{p}.$$

In this way we obtain an injective homomorphism

$$I_K \longrightarrow I_L ,$$

which will always be tacitly used to consider I_K as a subgroup of I_L. An element $\alpha = (\alpha_{\mathfrak{P}}) \in I_L$ therefore belongs to the group I_K if and only if its components $\alpha_{\mathfrak{P}}$ belong to $K_{\mathfrak{p}}$ $(\mathfrak{P}|\mathfrak{p})$, and if one has furthermore $\alpha_{\mathfrak{P}} = \alpha_{\mathfrak{P}'}$ whenever $\mathfrak{P}$ and $\mathfrak{P}'$ lie above the same place $\mathfrak{p}$ of K.

Every isomorphism $\sigma : L \to \sigma L$ induces an isomorphism

$$\sigma : I_L \longrightarrow I_{\sigma L}$$

like this. For each place $\mathfrak{P}$ of L, σ induces an isomorphism

$$\sigma : L_{\mathfrak{P}} \longrightarrow (\sigma L)_{\sigma\mathfrak{P}}.$$

For if we have $\alpha = \mathfrak{P}\text{-}\lim \alpha_i$, for some sequence $\alpha_i \in L$, then the sequence $\sigma\alpha_i \in \sigma L$ converges with respect to $|\ \ |_{\sigma\mathfrak{P}}$ in $(\sigma L)_{\sigma\mathfrak{P}}$, and the isomorphism is given by

$$\alpha = \mathfrak{P}\text{-}\lim \alpha_i \mapsto \sigma\alpha = \sigma\mathfrak{P}\text{-}\lim \sigma\alpha_i.$$

For an idèle $\alpha \in I_L$, we then define $\sigma\alpha \in I_L$ to be the idèle with components

$$(\sigma\alpha)_{\sigma\mathfrak{P}} = \sigma\alpha_{\mathfrak{P}} \in (\sigma L)_{\sigma\mathfrak{P}}.$$

If $L|K$ is a Galois extension with Galois group $G = G(L|K)$, then every $\sigma \in G$ yields an automorphism $\sigma : I_L \to I_L$, i.e., I_L is turned into an G-module. As to the fixed module $I_L^G = \{\alpha \in I_L \mid \sigma\alpha = \alpha \text{ for all } \sigma \in G\}$, we have the

(2.1) Proposition. *If $L|K$ is a Galois extension with Galois group G, then*

$$I_L^G = I_K.$$

Proof: Let $\alpha \in I_K \subseteq I_L$. For $\sigma \in G$, the induced map $\sigma : L_{\mathfrak{P}} \to L_{\sigma\mathfrak{P}}$ is a $K_{\mathfrak{p}}$-isomorphism, if $\mathfrak{P}|\mathfrak{p}$. Therefore

$$(\sigma\alpha)_{\sigma\mathfrak{P}} = \sigma\alpha_{\mathfrak{P}} = \alpha_{\mathfrak{P}} = \alpha_{\sigma\mathfrak{P}},$$

so that $\sigma\alpha = \alpha$, and therefore $\alpha \in I_L^G$. If conversely $\alpha = (\alpha_{\mathfrak{P}}) \in I_L^G$, then

$$(\sigma\alpha)_{\sigma\mathfrak{P}} = \sigma\alpha_{\mathfrak{P}} = \alpha_{\sigma\mathfrak{P}}$$

for all $\sigma \in G$. In particular, if σ belongs to the decomposition group $G_{\mathfrak{P}} = G(L_{\mathfrak{P}}|K_{\mathfrak{p}})$, then $\sigma\mathfrak{P} = \mathfrak{P}$ and $\sigma\alpha_{\mathfrak{P}} = \alpha_{\mathfrak{P}}$ so that $\alpha_{\mathfrak{P}} \in K_{\mathfrak{p}}^*$. If $\sigma \in G$ is arbitrary, then $\sigma : L_{\mathfrak{P}} \to L_{\sigma\mathfrak{P}}$ induces the identity on $K_{\mathfrak{p}}$, and we get $\alpha_{\mathfrak{P}} = \sigma\alpha_{\mathfrak{P}} = \alpha_{\sigma\mathfrak{P}}$ for any two places $\mathfrak{P}$ and $\sigma\mathfrak{P}$ above $\mathfrak{p}$. This shows that $\alpha \in I_K$. □

The idèle group I_L is the unit group of the ring of adèles $\mathbb{A}_L$ of L. It is convenient to write this ring as

$$\mathbb{A}_L = \prod_{\mathfrak{p}} L_{\mathfrak{p}},$$

where

$$L_{\mathfrak{p}} = \prod_{\mathfrak{P}|\mathfrak{p}} L_{\mathfrak{P}}.$$

The restricted product $\prod_{\mathfrak{p}} L_{\mathfrak{p}}$ consists of all families $(\alpha_{\mathfrak{p}})$ of elements $\alpha_{\mathfrak{p}} \in L_{\mathfrak{p}}$ such that $\alpha_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}} = \prod_{\mathfrak{P}|\mathfrak{p}} \mathcal{O}_{\mathfrak{P}}$ for almost all $\mathfrak{p}$. Via the diagonal embedding

$$K_{\mathfrak{p}} \longrightarrow L_{\mathfrak{p}},$$

the factor $L_{\mathfrak{p}}$ is a commutative $K_{\mathfrak{p}}$-algebra of degree $\sum_{\mathfrak{P}|\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [L : K]$. These embeddings yield the embedding

$$\mathbb{A}_K \longrightarrow \mathbb{A}_L,$$

whose restriction

$$I_K = \mathbb{A}_K^* \hookrightarrow \mathbb{A}_L^* = I_L$$

turns out to be the inclusion considered above.

Every $\alpha_{\mathfrak{p}} \in L_{\mathfrak{p}}^*$ defines an automorphism

$$\alpha_{\mathfrak{p}} : L_{\mathfrak{p}} \longrightarrow L_{\mathfrak{p}}, \quad x \longmapsto \alpha_{\mathfrak{p}} x,$$

of the $K_{\mathfrak{p}}$-vector space $L_{\mathfrak{p}}$, and as in the case of a field extension, we define the norm of $\alpha_{\mathfrak{p}}$ by

$$N_{L_{\mathfrak{p}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{p}}) = \det(\alpha_{\mathfrak{p}}).$$

In this way we obtain a homomorphism

$$N_{L_{\mathfrak{p}}|K_{\mathfrak{p}}} : L_{\mathfrak{p}}^* \longrightarrow K_{\mathfrak{p}}^*.$$

It induces a norm homomorphism

$$N_{L|K} : I_L \longrightarrow I_K$$

between the idèle groups $I_L = \prod_{\mathfrak{p}} L_{\mathfrak{p}}^*$ and $I_K = \prod_{\mathfrak{p}} K_{\mathfrak{p}}^*$. Explicitly the norm of an idèle is given by the following proposition.

(2.2) Proposition. *If $L|K$ is a finite extension and $\alpha = (\alpha_{\mathfrak{P}}) \in I_L$, the local components of the idèle $N_{L|K}(\alpha)$ are given by*

$$N_{L|K}(\alpha)_{\mathfrak{p}} = \prod_{\mathfrak{P}|\mathfrak{p}} N_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{P}}).$$

Proof: Putting $\alpha_{\mathfrak{p}} = (\alpha_{\mathfrak{P}})_{\mathfrak{P}|\mathfrak{p}} \in L_{\mathfrak{p}}$, the $K_{\mathfrak{p}}$-automorphism $\alpha_{\mathfrak{p}} : L_{\mathfrak{p}} \to L_{\mathfrak{p}}$ is the direct product of the $K_{\mathfrak{p}}$-automorphisms $\alpha_{\mathfrak{P}} : L_{\mathfrak{P}} \to L_{\mathfrak{P}}$. Therefore

$$N_{L_{\mathfrak{p}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{p}}) = \det(\alpha_{\mathfrak{p}}) = \prod_{\mathfrak{P}|\mathfrak{p}} \det(\alpha_{\mathfrak{P}}) = \prod_{\mathfrak{P}|\mathfrak{p}} N_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{P}}). \qquad \square$$

The idèle norm enjoys the following properties.

(2.3) Proposition. (i) *For a tower of fields* $K \subseteq L \subseteq M$ *we have* $N_{M|K} = N_{L|K} \circ N_{M|L}$.

(ii) *If* $L|K$ *is embedded into the Galois extension* $M|K$ *and if* $G = G(M|K)$ *and* $H = G(M|L)$, *then one has for* $\alpha \in I_L$: $N_{L|K}(\alpha) = \prod_{\sigma \in G/H} \sigma\alpha$.

(iii) $N_{L|K}(\alpha) = \alpha^{[L:K]}$ *for* $\alpha \in I_K$.

(iv) *The norm of the principal idèle* $x \in L^*$ *is the principal idèle of* K *defined by the usual norm* $N_{L|K}(x)$.

The proofs of (i), (ii), (iii) are literally the same as for the norm in a field extension (see chap. I, §2). (iv) follows from the fact that, once we identify $L_\mathfrak{p} = L \otimes_K K_\mathfrak{p}$ (see chap. II, (8.3)), the $K_\mathfrak{p}$-automorphism $f_x : L_\mathfrak{p} \to L_\mathfrak{p}$, $y \mapsto xy$, arises from the K-automorphism $x : L \to L$ by tensoring with $K_\mathfrak{p}$. Hence $\det(f_x) = \det(x)$.

Remark: For fundamental as well as practical reasons, it is convenient to adopt a formal point of view for the above considerations which allows us to avoid the constant back and forth between idèles and their components. This point of view is based on identifying the ring of adèles $\mathbb{A}_L$ of L as

$$\mathbb{A}_L = \mathbb{A}_K \otimes_K L,$$

which results from the canonical isomorphisms (see chap. II, (8.3))

$$K_\mathfrak{p} \otimes_K L \xrightarrow{\sim} L_\mathfrak{p} = \prod_{\mathfrak{P}|\mathfrak{p}} L_\mathfrak{P}, \quad \alpha_\mathfrak{p} \otimes a \longmapsto \alpha_\mathfrak{p} \cdot (\tau_\mathfrak{P} a).$$

Here $\tau_\mathfrak{P}$ denotes the canonical embedding $\tau_\mathfrak{P} : L \to L_\mathfrak{P}$.

In this way the inclusion by components $I_K \subseteq I_L$ is simply given by the embedding $\mathbb{A}_K \hookrightarrow \mathbb{A}_L$, $\alpha \mapsto \alpha \otimes 1$, induced by $K \subseteq L$. An isomorphism $L \to \sigma L$ then yields the isomorphism

$$\sigma : \mathbb{A}_L = \mathbb{A}_K \otimes_K L \longrightarrow \mathbb{A}_K \otimes_K \sigma L = \mathbb{A}_{\sigma L}$$

via $\sigma(\alpha \otimes a) = \alpha \otimes \sigma a$, and the norm of an L-idèle $\alpha \in \mathbb{A}_L^*$ is simply the determinant

$$N_{L|K}(\alpha) = \det_{\mathbb{A}_K}(\alpha)$$

of the endomorphism $\alpha : \mathbb{A}_L \to \mathbb{A}_L$ which α induces on the finite $\mathbb{A}_K$-algebra $\mathbb{A}_L = \mathbb{A}_K \otimes_K L$.

Here are consequences of the preceding investigations for the **idèle class groups**.

(2.4) Proposition. *If $L|K$ is a finite extension, then the homomorphism $I_K \to I_L$ induces an injection of idèle class groups*

$$C_K \longrightarrow C_L\,, \quad \alpha K^* \longmapsto \alpha L^*\,.$$

Proof: The injection $I_K \to I_L$ clearly maps K^* into L^*. For the injectivity, we have to show that $I_K \cap L^* = K^*$. Let $M|K$ be a finite Galois extension with Galois group G containing L. Then we have $I_K \subseteq I_L \subseteq I_M$, and

$$I_K \cap L^* \subseteq I_K \cap M^* \subseteq (I_K \cap M^*)^G = I_K \cap M^{*G} = I_K \cap K^* = K^*\,. \quad \square$$

Via the embedding $C_K \to C_L$, the idèle class group C_K becomes a subgroup of C_L: an element $\alpha L^* \in C_L$ ($\alpha \in I_L$) lies in C_K if and only if the class αL^* has a representative α' in I_K. It is important to know that we have **Galois descent** for the idèle class group:

(2.5) Proposition. *If $L|K$ is a Galois extension and $G = G(L|K)$, then C_L is canonically a G-module and $C_L^G = C_K$.*

Proof: The G-module I_L contains L^* as a G-submodule. Hence every $\sigma \in G$ induces an automorphism

$$C_L \xrightarrow{\sigma} C_L\,, \quad \alpha L^* \longmapsto (\sigma\alpha)L^*\,.$$

This gives us an exact sequence of G-modules

$$1 \longrightarrow L^* \longrightarrow I_L \longrightarrow C_L \longrightarrow 1\,.$$

We claim that the sequence

$$1 \longrightarrow L^{*G} \longrightarrow I_L^G \longrightarrow C_L^G \longrightarrow 1$$

deduced from the first is still exact. The injectivity of $L^{*G} \to I_L^G$ is trivial. The kernel of $I_L^G \to C_L^G$ is $I_L^G \cap L^* = I_K \cap L^* = K^* = L^{*G}$. The surjectivity of $I_L^G \to C_L^G$ is not altogether straightforward. To prove it, let $\alpha L^* \in C_L^G$. For every $\sigma \in G$, one then has $\sigma(\alpha L^*) = \alpha L^*$, i.e., $\sigma\alpha = \alpha x_\sigma$ for some $x_\sigma \in L^*$. This x_σ is a "crossed homomorphism", i.e., we have

$$x_{\sigma\tau} = x_\sigma \cdot \sigma x_\tau\,.$$

Indeed, $x_{\sigma\tau} = \frac{\sigma\tau\alpha}{\alpha} = \frac{\sigma\tau\alpha}{\sigma\alpha}\cdot\frac{\sigma\alpha}{\alpha} = \sigma(\frac{\tau\alpha}{\alpha})\frac{\sigma\alpha}{\alpha} = \sigma x_\tau x_\sigma$. By Hilbert 90 in Noether's version (see chap. IV, (3.8)) such a crossed homomorphism is of the form $x_\sigma = \sigma y/y$ for some $y \in L^*$. Putting $\alpha' = \alpha y^{-1}$ yields $\alpha' L^* = \alpha L^*$ and $\sigma\alpha' = \sigma\alpha\sigma y^{-1} = \alpha x_\sigma \sigma y^{-1} = \alpha y^{-1} = \alpha'$, hence $\alpha' \in I_L^G$. This proves surjectivity. $\square$

The norm map $N_{L|K} : I_L \to I_K$ sends principal idèles to principal idèles by (2.3). Hence we get a norm map also for the idèle class group C_L,

$$N_{L|K} : C_L \longrightarrow C_K .$$

It enjoys the same properties (2.3), (i), (ii), (iii), as the norm map on the idèle group.

Exercise 1. Let $\omega_1, \dots, \omega_n$ be a basis of $L|K$. Then the isomorphism $L \otimes_K K_\mathfrak{p} \cong \prod_{\mathfrak{P}|\mathfrak{p}} L_\mathfrak{P}$ induces, for almost all prime ideals $\mathfrak{p}$ of K, an isomorphism

$$\omega_1 \mathcal{O}_\mathfrak{p} \oplus \cdots \oplus \omega_n \mathcal{O}_\mathfrak{p} \cong \prod_{\mathfrak{P}|\mathfrak{p}} \mathcal{O}_\mathfrak{P},$$

where $\mathcal{O}_\mathfrak{p}$, resp. $\mathcal{O}_\mathfrak{P}$, is the valuation ring of $K_\mathfrak{p}$, resp. $L_\mathfrak{P}$.

Exercise 2. Let $L|K$ be a finite extension. The absolute norm $\mathfrak{N}$ of idèles of K, resp. L, behaves as follows under the inclusion $i_{L|K} : I_K \to I_L$, resp. under the norm $N_{L|K} : I_L \to I_K$:

$$\mathfrak{N}(i_{L|K}(\alpha)) = \mathfrak{N}(\alpha)^{[L:K]} \quad \text{for} \quad \alpha \in I_K ,$$
$$\mathfrak{N}(N_{L|K}(\alpha)) = \mathfrak{N}(\alpha) \quad \text{for} \quad \alpha \in I_L .$$

Exercise 3. The correspondence between idèles and ideals, $\alpha \mapsto (\alpha)$, satisfies the following rule, in the case of a Galois extension $L|K$,

$$(N_{L|K}(\alpha)) = N_{L|K}((\alpha)) .$$

(For the norm on ideals, see chap. III, § 1.)

Exercise 4. The ideal class group, unlike the idèle class group, does not have Galois descent. More precisely, for a Galois extension $L|K$, the homomorphism $Cl_K \to Cl_L^{G(L|K)}$ is in general neither injective nor surjective.

Exercise 5. Define the trace $Tr_{L|K} : \mathbb{A}_L \to \mathbb{A}_K$ by $Tr_{L|K}(\alpha) =$ trace of the endomorphism $x \mapsto \alpha x$ of the $\mathbb{A}_K$-algebra $\mathbb{A}_L$, and show:

(i) $Tr_{L|K}(\alpha)_\mathfrak{p} = \sum_{\mathfrak{P}|\mathfrak{p}} Tr_{L_\mathfrak{P}|K_\mathfrak{p}}(\alpha_\mathfrak{P})$.

(ii) For a tower of fields $K \subseteq L \subseteq M$, one has $Tr_{M|K} = Tr_{L|K} \circ Tr_{M|L}$.

(iii) If $L|K$ is embedded into the Galois extension $M|K$, and if $G = G(M|K)$ and $H = G(M|L)$, then one has for $\alpha \in \mathbb{A}_L$, $Tr_{L|K}(\alpha) = \sum_{\sigma \in G/H} \sigma\alpha$.

(iv) $Tr_{L|K}(\alpha) = [L : K]\alpha \quad$ for $\alpha \in \mathbb{A}_K$.

(v) The trace of a principal adèle $x \in L$ is the principal adèle in $\mathbb{A}_K$ defined by the usual trace $Tr_{L|K}(x)$.

§ 3. The Herbrand Quotient of the Idèle Class Group

Our goal now is to show that the idèle class group satisfies the class field axiom of chap. IV, (6.1). To do this we will first compute its Herbrand quotient. It is constituted on the one hand by the Herbrand quotient of the idèle group, and by that of the unit group on the other. We study the idèle group first.

Let $L|K$ be a finite Galois extension with Galois group G. The G-module I_L may be described in the following simple manner, which immediately reduces us to local fields. For every place $\mathfrak{p}$ of K we put

$$L_{\mathfrak{p}}^* = \prod_{\mathfrak{P}|\mathfrak{p}} L_{\mathfrak{P}}^* \quad \text{and} \quad U_{L,\mathfrak{p}} = \prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}} .$$

Since the automorphisms $\sigma \in G$ permute the places of L above $\mathfrak{p}$, the groups $L_{\mathfrak{p}}^*$ and $U_{L,\mathfrak{p}}$ are G-modules, and we have for the G-module I_L the decomposition

$$I_L = \prod_{\mathfrak{p}} L_{\mathfrak{p}}^* ,$$

where the restricted product is taken with respect to the subgroups $U_{L,\mathfrak{p}} \subseteq L_{\mathfrak{p}}^*$. Choose a place $\mathfrak{P}$ of L above $\mathfrak{p}$, and let $G_{\mathfrak{P}} = G(L_{\mathfrak{P}}|K_{\mathfrak{p}}) \subseteq G$ be its decomposition group. As σ varies over a system of representatives of $G/G_{\mathfrak{P}}$, $\sigma\mathfrak{P}$ runs through the various places of L above $\mathfrak{p}$, and we get

$$L_{\mathfrak{p}}^* = \prod_{\sigma} L_{\sigma\mathfrak{P}}^* = \prod_{\sigma} \sigma(L_{\mathfrak{P}}^*) , \quad U_{L,\mathfrak{p}} = \prod_{\sigma} U_{\sigma\mathfrak{P}} = \prod_{\sigma} \sigma(U_{\mathfrak{P}}) .$$

In terms of the notion of *induced module* introduced in chap. IV, §7, we thus get the following

(3.1) Proposition. *$L_{\mathfrak{p}}^*$ and $U_{L,\mathfrak{p}}$ are the induced G-modules*

$$L_{\mathfrak{p}}^* = \operatorname{Ind}_G^{G_{\mathfrak{P}}}(L_{\mathfrak{P}}^*) , \quad U_{L,\mathfrak{p}} = \operatorname{Ind}_G^{G_{\mathfrak{P}}}(U_{\mathfrak{P}}) .$$

Now let S be a finite set of places of K containing the infinite places. We then define $I_L^S = I_L^{\bar{S}}$, where $\bar{S}$ denotes the set of all places of L which lie above the places of S. For I_L^S we have the G-module decomposition

$$I_L^S = \prod_{\mathfrak{p}\in S} L_{\mathfrak{p}}^* \times \prod_{\mathfrak{p}\notin S} U_{L,\mathfrak{p}} ,$$

and (3.1) gives the

(3.2) Proposition. *If $L|K$ is a cyclic extension, and if S contains all primes ramified in L, then we have for $i = 0,\ -1$ that*

$$H^i(G, I_L^S) \cong \bigoplus_{\mathfrak{p}\in S} H^i(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*) \quad \text{and} \quad H^i(G, I_L) \cong \bigoplus_{\mathfrak{p}} H^i(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*) ,$$

where for each $\mathfrak{p}$, $\mathfrak{P}$ is a chosen prime of L above $\mathfrak{p}$.

Proof: The decomposition $I_L^S = (\bigoplus_{\mathfrak{p}\in S} L_\mathfrak{p}^*) \oplus V$, $V = \prod_{\mathfrak{p}\notin S} U_{L,\mathfrak{p}}$, gives us an isomorphism

$$H^i(G, I_L^S) = \bigoplus_{\mathfrak{p}\in S} H^i(G, L_\mathfrak{p}^*) \;\oplus\; H^i(G, V),$$

and an injection $H^i(G,V) \to \prod_{\mathfrak{p}\notin S} H^i(G, U_{L,\mathfrak{p}})$. By (3.1) and chap. IV, (7.4), we have the isomorphisms $H^i(G, L_\mathfrak{p}^*) \cong H^i(G_\mathfrak{P}, L_\mathfrak{P}^*)$ and $H^i(G, U_{L,\mathfrak{p}}) \cong H^i(G_\mathfrak{P}, U_\mathfrak{P})$. For $\mathfrak{p} \notin S$, $L_\mathfrak{P}|K_\mathfrak{p}$ is unramified. Hence $H^i(G_\mathfrak{P}, U_\mathfrak{P}) = 1$, by chap. V, (1.2). This shows the first claim of the proposition. The second is an immediate consequence:

$$H^i(G, I_L) = \varinjlim_S H^i(G, I_L^S) \cong \varinjlim_S \bigoplus_{\mathfrak{p}\in S} H^i(G_\mathfrak{P}, L_\mathfrak{P}^*) = \bigoplus_{\mathfrak{p}} H^i(G_\mathfrak{P}, L_\mathfrak{P}^*).$$

□

The proposition says that one has $H^{-1}(G, I_L) = \{1\}$, because $H^{-1}(G_\mathfrak{P}, L_\mathfrak{P}^*) = \{1\}$ by Hilbert 90. Further it says that

$$I_K / N_{L|K} I_L = \bigoplus_{\mathfrak{p}} K_\mathfrak{p}^* / N_{L_\mathfrak{P}|K_\mathfrak{p}} L_\mathfrak{P}^*,$$

where $\mathfrak{P}$ is a chosen place above $\mathfrak{p}$. In other words:

An idèle $\alpha \in I_K$ is a norm of an idèle of L if and only if it is *a norm locally everywhere*, i.e., if every component $\alpha_\mathfrak{p}$ is the norm of an element of $L_\mathfrak{P}^*$.

As for the Herbrand quotient $h(G, I_L^S)$ we obtain the result:

(3.3) Proposition. *If $L|K$ is a cyclic extension and if S contains all ramified primes, then*

$$h(G, I_L^S) = \prod_{\mathfrak{p}\in S} n_\mathfrak{p},$$

where $n_\mathfrak{p} = [L_\mathfrak{P} : K_\mathfrak{p}]$.

Proof: We have $H^{-1}(G, I_L^S) = \prod_{\mathfrak{p}\in S} H^{-1}(G_\mathfrak{P}, L_\mathfrak{P}^*) = 1$ and

$$H^0(G, I_L^S) = \prod_{\mathfrak{p}\in S} H^0(G_\mathfrak{P}, L_\mathfrak{P}^*).$$

By local class field theory, we find $\#H^0(G_\mathfrak{P}, L_\mathfrak{P}^*) = (K_\mathfrak{p}^* : N_{L_\mathfrak{P}|K_\mathfrak{p}} L_\mathfrak{P}^*) = n_\mathfrak{p}$. Hence

$$h(G, I_L^S) = \frac{\#H^0(G, I_L^S)}{\#H^{-1}(G, I_L^S)} = \prod_{\mathfrak{p}\in S} n_\mathfrak{p}.$$

□

Next we determine the Herbrand quotient of the G-module $L^S = L \cap I_L^S$. For this we need the following general

(3.4) Lemma. *Let V be an s-dimensional $\mathbb{R}$-vector space, and let G be a finite group of automorphisms of V which operates as a permutation group on the elements of a basis $v_1, \dots, v_s$: $\sigma v_i = v_{\sigma(i)}$.*

If Γ is a G-invariant complete lattice in V, i.e., $\sigma\Gamma \subseteq \Gamma$ for all σ, then there exists a complete sublattice in Γ,

$$\Gamma' = \mathbb{Z}w_1 + \cdots + \mathbb{Z}w_s ,$$

such that $\sigma w_i = w_{\sigma(i)}$ for all $\sigma \in G$.

Proof: Let $|\ |$ be the sup-norm with respect to the coordinates of the basis $v_1, \dots, v_s$. Since Γ is a lattice, there exists a number b such that for every $x \in V$, there is a $\gamma \in \Gamma$ satisfying

$$|x - \gamma| < b .$$

Choose a large positive number $t \in \mathbb{R}$, and a $\gamma \in \Gamma$ such that

$$|tv_1 - \gamma| < b,$$

and define

$$w_i = \sum_{\sigma(1)=i} \sigma\gamma , \quad i = 1, \dots, s ,$$

i.e., the summation is over all $\sigma \in G$ such that $\sigma(1) = i$. For every $\tau \in G$ we then have

$$\tau w_i = \sum_{\sigma(1)=i} \tau\sigma\gamma = \sum_{\rho(1)=\tau(i)} \rho\gamma = w_{\tau(i)} .$$

It is therefore enough to check the linear independence of the w_i. To do this, let

$$\sum_{i=1}^{s} c_i w_i = 0, \quad c_i \in \mathbb{R} .$$

If not all of the $c_i = 0$, then we may assume $|c_i| \le 1$ and $c_j = 1$ for some j. Let

$$\gamma = tv_1 - y,$$

for some vector y of absolute value $|y| < b$. Then

$$w_i = \sum_{\sigma(1)=i} \sigma\gamma = t \sum_{\sigma(1)=i} v_{\sigma(1)} - y_i = tn_i v_i - y_i ,$$

where $|y_i| \le gb$, for $g = \#G$, and $n_i = \#\{\sigma \in G \mid \sigma(1) = i\}$. We therefore get

$$0 = \sum_{i=1}^{s} c_i w_i = t \sum_{i=1}^{s} c_i n_i v_i - z ,$$

with $|z| \le sgb$, i.e.,

$$z = tn_j v_j + \sum_{i \ne j} tc_i n_i v_i .$$

If t was chosen sufficiently large, then z cannot be written in this way. This contradiction proves the lemma. $\square$

Now let $L|K$ be a cyclic extension of degree n with Galois group $G = G(L|K)$, let S be a finite set of places containing the infinite places, and let $\bar{S}$ be the set of places of L that lie above the places of S. We denote the group $L^{\bar{S}}$ of $\bar{S}$-units simply by L^S.

(3.5) Proposition. *The Herbrand quotient of the G-module L^S satisfies*

$$h(G, L^S) = \frac{1}{n} \prod_{\mathfrak{p} \in S} n_{\mathfrak{p}},$$

where $n_{\mathfrak{p}} = [L_{\mathfrak{P}} : K_{\mathfrak{p}}]$.

Proof: Let $\{\mathfrak{e}_{\mathfrak{P}} \mid \mathfrak{P} \in \bar{S}\}$ be the standard basis of the vector space $V = \prod_{\mathfrak{P} \in \bar{S}} \mathbb{R}$. By (1.1), the homomorphism

$$\lambda : L^S \longrightarrow V, \quad \lambda(a) = \sum_{\mathfrak{P} \in \bar{S}} \log |a|_{\mathfrak{P}} \mathfrak{e}_{\mathfrak{P}},$$

has kernel $\mu(L)$ and its image is an $(\bar{s} - 1)$-dimensional lattice, $\bar{s} = \#\bar{S}$. We make G operate on V via

$$\sigma \mathfrak{e}_{\mathfrak{P}} = \mathfrak{e}_{\sigma\mathfrak{P}}.$$

Then λ is a G-homomorphism because we have, for $\sigma \in G$,

$$\begin{aligned}\lambda(\sigma a) &= \sum_{\mathfrak{P}} \log |\sigma a|_{\mathfrak{P}} \mathfrak{e}_{\mathfrak{P}} = \sum_{\mathfrak{P}} \log |a|_{\sigma^{-1}\mathfrak{P}} \sigma \mathfrak{e}_{\sigma^{-1}\mathfrak{P}} \\ &= \sigma\Big(\sum_{\mathfrak{P}} \log |a|_{\sigma^{-1}\mathfrak{P}} \mathfrak{e}_{\sigma^{-1}\mathfrak{P}}\Big) = \sigma\lambda(a).\end{aligned}$$

Therefore $\mathfrak{e}_0 = \sum_{\mathfrak{P} \in \bar{S}} \mathfrak{e}_{\mathfrak{P}}$ and $\lambda(L^S)$ generate a G-invariant complete lattice Γ in V. Since $\mathbb{Z}\mathfrak{e}_0$ is G-isomorphic to $\mathbb{Z}$, the exact sequence

$$0 \longrightarrow \mathbb{Z}\mathfrak{e}_0 \longrightarrow \Gamma \longrightarrow \Gamma/\mathbb{Z}\mathfrak{e}_0 \longrightarrow 0,$$

together with the fact that $\Gamma/\mathbb{Z}\mathfrak{e}_0 = \lambda(L^S)$, yields the identities

$$h(G, L^S) = h(G, \lambda(L^S)) = h(G, \mathbb{Z})^{-1} h(G, \Gamma) = \frac{1}{n} h(G, \Gamma).$$

We now choose in Γ a sublattice Γ', in accordance with lemma (3.4). Then we have

$$\Gamma' = \bigoplus_{\mathfrak{P}} \mathbb{Z} w_{\mathfrak{P}} = \bigoplus_{\mathfrak{p} \in S} \bigoplus_{\mathfrak{P} | \mathfrak{p}} \mathbb{Z} w_{\mathfrak{P}} = \bigoplus_{\mathfrak{p} \in S} \Gamma'_{\mathfrak{p}}$$

and $\sigma w_{\mathfrak{P}} = w_{\sigma\mathfrak{P}}$. This identifies $\Gamma'_{\mathfrak{p}}$ as the induced G-module

$$\Gamma'_{\mathfrak{p}} = \bigoplus_{\mathfrak{P} | \mathfrak{p}} \mathbb{Z} w_{\mathfrak{P}} = \bigoplus_{\sigma \in G/G_{\mathfrak{p}}} \sigma(\mathbb{Z} w_{\mathfrak{P}_0}) = \operatorname{Ind}_G^{G_{\mathfrak{p}}}(\mathbb{Z} w_{\mathfrak{P}_0}),$$

where $\mathfrak{P}_0$ is a chosen place above $\mathfrak{p}$, and $G_\mathfrak{p}$ is its decomposition group. The lattice Γ' has the same rank as Γ, so is therefore of finite index in Γ. From chap. IV, (7.4), we conclude that

$$\begin{aligned} h(G,L^S) &= \frac{1}{n}h(G,\Gamma') = \frac{1}{n}\prod_{\mathfrak{p}\in S} h(G,\Gamma'_\mathfrak{p}) = \frac{1}{n}\prod_{\mathfrak{p}\in S} h(G_\mathfrak{p},\mathbb{Z}w_{\mathfrak{P}_0}) \\ &= \frac{1}{n}\prod_{\mathfrak{p}\in S} h(G_\mathfrak{p},\mathbb{Z}). \end{aligned}$$

Thus we do find that $h(G,L^S) = \frac{1}{n}\prod_{\mathfrak{p}\in S} n_\mathfrak{p}$, where $n_\mathfrak{p} = \#G_\mathfrak{p} = [L_\mathfrak{P} : K_\mathfrak{p}]$. $\square$

From the Herbrand quotient of I_L^S and L^S we immediately get the Herbrand quotient of the idèle class group C_L. To do it choose a finite set of places S containing all infinite ones and all primes ramified in L, such that $I_L = I_L^S L^*$. Such a set exists by (1.4). From the exact sequence

$$1 \longrightarrow L^S \longrightarrow I_L^S \longrightarrow I_L^S L^*/L^* \longrightarrow 1$$

arises the identity

$$h(G,C_L) = h(G,I_L^S)h(G,L^S)^{-1},$$

and from (3.3) and (3.5) we obtain the

(3.6) Theorem. *If $L|K$ is a cyclic extension of degree n with Galois group $G = G(L|K)$, then*

$$h(G,C_L) = \frac{\#H^0(G,C_L)}{\#H^{-1}(G,C_L)} = n.$$

In particular $(C_K : N_{L|K}C_L) \geq n$.

From this result we deduce the following interesting consequence.

(3.7) Corollary. *If $L|K$ is cyclic of prime power degree $n = p^\nu$ $(\nu > 0)$, then there are infinitely many places of K which do not split in L.*

Proof: Assume that the set S of nonsplit primes were finite. Let $M|K$ be the subextension of $L|K$ of degree p. For every $\mathfrak{p} \notin S$, the decomposition group $G_\mathfrak{p}$ of $L|K$ is different from $G(L|K)$. Hence $G_\mathfrak{p} \subseteq G(L|M)$. Therefore every $\mathfrak{p} \notin S$ splits completely in M. We deduce from this that $N_{M|K}C_M = C_K$, thus contradicting (3.6).

Indeed, let $\alpha \in I_K$. By the approximation theorem of chap. II, (3.4), there exists an $a \in K^*$ such that $\alpha_{\mathfrak{p}}a^{-1}$ is contained in the open subgroup $N_{M_{\mathfrak{P}}|K_{\mathfrak{p}}}M_{\mathfrak{P}}^*$, for all $\mathfrak{p} \in S$. If $\mathfrak{p} \notin S$, then $\alpha_{\mathfrak{p}}a^{-1}$ is automatically contained in $N_{M_{\mathfrak{P}}|K_{\mathfrak{p}}}M_{\mathfrak{P}}^*$ because $M_{\mathfrak{P}} = K_{\mathfrak{p}}$. Since

$$I_K/N_{M|K}I_M = \bigoplus_{\mathfrak{p}} K_{\mathfrak{p}}^*/N_{M_{\mathfrak{P}}|K_{\mathfrak{p}}}M_{\mathfrak{P}}^*,$$

the idèle αa^{-1} is a norm of some idèle β of I_M, i.e., $\alpha = (N_{M|K}\beta)a \in N_{M|K}I_MK^*$. This shows that the class of α belongs to $N_{M|K}C_M$, so that $C_K = N_{M|K}C_M$. □

(3.8) Corollary. *Let $L|K$ be a finite extension of algebraic number fields. If almost all primes of K split completely in L, then $L = K$.*

Proof: We may assume without loss of generality that $L|K$ is Galois. In fact, let $M|K$ be the normal closure of $L|K$, and write $G = G(M|K)$ and $H = G(M|L)$. Also let $\mathfrak{p}$ be a place of K, $\mathfrak{P}$ a place of M above $\mathfrak{p}$, and let $G_{\mathfrak{P}}$ be its decomposition group. Then the number of places of L above $\mathfrak{p}$ equals the number $\#H\backslash G/G_{\mathfrak{P}}$ of double cosets $H\sigma G_{\mathfrak{P}}$ in G (see chap. I, §9). Hence $\mathfrak{p}$ splits completely in L if $\#H\backslash G/G_{\mathfrak{P}} = [L:K] = \#H\backslash G$. But this is tantamount to $G_{\mathfrak{P}} = 1$, and hence to the fact that $\mathfrak{p}$ splits completely in M.

So assume $L|K$ is Galois, $L \neq K$, and let $\sigma \in G(L|K)$ be an element of prime order, with fixed field K'. If almost all primes $\mathfrak{p}$ of K were completely split in L, then the same would hold for the primes $\mathfrak{p}'$ of K'. This contradicts (3.7). □

Exercise 1. If the Galois extension $L|K$ is not cyclic, then there are at most finitely many primes of K which do not split in L.

Exercise 2. If $L|K$ is a finite Galois extension, then the Galois group $G(L|K)$ is generated by the Frobenius automorphisms $\varphi_{\mathfrak{P}}$ of all prime ideals $\mathfrak{P}$ of L which are unramified over K.

Exercise 3. Let $L|K$ be a finite abelian extension, and let D be a subgroup of I_K such that K^*D is dense in I_K and $D \subseteq N_{L|K}L^*$. Then $L = K$.

Exercise 4. Let $L_1, \dots, L_r|K$ be cyclic extensions of prime degree p such that $L_i \cap L_j = K$ for $i \neq j$. Then there are infinitely many primes $\mathfrak{p}$ of K which split completely in L_i, for $i > 1$, but which are nonsplit in L_1.

§ 4. The Class Field Axiom

Having determined the Herbrand quotient $h(G,C_L)$ to be the degree $n=[L:K]$ of the cyclic extension $L|K$, it will now be enough to show either $H^{-1}(G,C_L)=1$ or $H^0(G,C_L)=(C_K:N_{L|K}C_L)=n$. The first identity is curiously inaccessible by way of direct attack. We are thus stuck with the second. We will reduce the problem to the case of a *Kummer extension.* For such an extension the norm group $N_{L|K}C_L$ can be written down explicitly, and this allows us to compute the index $(C_K:N_{L|K}C_L)$.

So let K be a number field that contains the n-th roots of unity, where n is a fixed *prime power*, and let $L|K$ be a Galois extension with a Galois group of the form

$$G(L|K)\cong(\mathbb{Z}/n\mathbb{Z})^r.$$

We choose a finite set of places S containing the ramified places, those that divide n, and the infinite ones, and which is such that $I_K=I_K^SK^*$. We write again $K^S=I_K^S\cap K^*$ for the group of S-units, and we put $s=\#S$.

(4.1) Proposition. *One has $s\geq r$, and there exists a set T of $s-r$ primes of K that do not belong to S such that*

$$L=K(\sqrt[n]{\Delta}),$$

where Δ is the kernel of the map $K^S\to\prod_{\mathfrak{p}\in T}K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n}$.

Proof: We show first that $L=K(\sqrt[n]{\Delta})$ if $\Delta=L^{*n}\cap K^S$, and then that Δ is the said kernel. By chap. IV, (3.6), we certainly have that $L=K(\sqrt[n]{D})$, with $D=L^{*n}\cap K$. If $x\in D$, then $K_{\mathfrak{p}}(\sqrt[n]{x})|K_{\mathfrak{p}}$ is unramified for all $\mathfrak{p}\notin S$ because S contains the places ramified in L. By chap. V, (3.3), we may therefore write $x=u_{\mathfrak{p}}y_{\mathfrak{p}}^n$, with $u_{\mathfrak{p}}\in U_{\mathfrak{p}}$, $y_{\mathfrak{p}}\in K_{\mathfrak{p}}^*$. Putting $y_{\mathfrak{p}}=1$ for $\mathfrak{p}\in S$, we get an idèle $y=(y_{\mathfrak{p}})$ which can be written as a product $y=\alpha z$ with $\alpha\in I_K^S$, $z\in K^*$. Then $xz^{-n}=u_{\mathfrak{p}}\alpha_{\mathfrak{p}}^n\in U_{\mathfrak{p}}$ for all $\mathfrak{p}\notin S$, i.e., $xz^{-n}\in I_K^S\cap K^*=K^S$, so that $xz^{-n}\in\Delta$. This shows that $D=\Delta K^{*n}$, and thus $L=K(\sqrt[n]{\Delta})$.

The field $N=K(\sqrt[n]{K^S})$ contains the field L because $\Delta=L^{*n}\cap K^S\subseteq K^S$. By Kummer theory, chap. IV, (3.6), we have

$$G(N|K)\cong\operatorname{Hom}\big(K^S/(K^S)^n,\mathbb{Z}/n\mathbb{Z}\big).$$

By (1.1), K^S is the product of a free group of rank $s-1$ and of the cyclic group $\mu(K)$ whose order is divisible by n. Therefore $K^S/(K^S)^n$

is a free $(\mathbb{Z}/n\mathbb{Z})$-module of rank s, and so is $G(N|K)$. Moreover, $G(N|K)/G(N|L) \cong G(L|K) \cong (\mathbb{Z}/n\mathbb{Z})^r$ is a free $(\mathbb{Z}/n\mathbb{Z})$-module of rank r so that $r \leq s$, and $G(N|L)$ is a free $(\mathbb{Z}/n\mathbb{Z})$-module of rank $s-r$. Let $\sigma_1, \dots, \sigma_{s-r}$ be a $\mathbb{Z}/n\mathbb{Z}$-basis of $G(N|L)$, and let N_i be the fixed field of σ_i, $i = 1, \dots, s-r$. Then $L = \bigcap_{i=1}^{s-r} N_i$. For every $i = 1, \dots, s-r$ we choose a prime $\mathfrak{P}_i$ of N_i which is nonsplit in N such that the primes $\mathfrak{p}_1, \dots, \mathfrak{p}_{s-r}$ of K lying below $\mathfrak{P}_1, \dots, \mathfrak{P}_{s-r}$ are all distinct, and do not belong to S. This is possible by (3.7). We now show that the set $T = \{\mathfrak{p}_1, \dots, \mathfrak{p}_{s-r}\}$ realizes the group $\Delta = L^{*n} \cap K^S$ as the kernel of $K^S \to \prod_{\mathfrak{p}\in T} K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n}$.

N_i is the decomposition field of $N|K$ at the unique prime $\mathfrak{P}_i'$ above $\mathfrak{P}_i$, for $i = 1, \dots, s-r$. Indeed, this decomposition field Z_i is contained in N_i because $\mathfrak{P}_i$ is nonsplit in N. On the other hand, the prime $\mathfrak{p}_i$ is unramified in N, because by chap. V, (3.3), it is unramified in every extension $K(\sqrt[n]{u})$, $u \in K^S$. The decomposition group $G(N|Z_i) \supseteq G(N|N_i)$ is therefore cyclic, and necessarily of order n since each element of $G(N|K)$ has order dividing n. This shows that $N_i = Z_i$.

From $L = \bigcap_{i=1}^{s-r} N_i$ it follows that $L|K$ is the maximal subextension of $N|K$ in which the primes $\mathfrak{p}_1, \dots, \mathfrak{p}_{s-r}$ split completely. For $x \in K^S$ we therefore have

$$\begin{aligned} x \in \Delta \iff K(\sqrt[n]{x}) \subseteq L &\iff K_{\mathfrak{p}_i}(\sqrt[n]{x}) = K_{\mathfrak{p}_i},\ i = 1, \dots, s-r, \\ &\iff x \in K_{\mathfrak{p}_i}^{*n},\ i = 1, \dots, s-r. \end{aligned}$$

This shows that Δ is the kernel of the map $K^S \to \prod_{i=1}^{s-r} K_{\mathfrak{p}_i}^*/K_{\mathfrak{p}_i}^{*n}$. □

(4.2) Theorem. *Let T be a set of places as in* (4.1), *and let*

$$C_K(S,T) = I_K(S,T)K^*/K^*,$$

where

$$I_K(S,T) = \prod_{\mathfrak{p}\in S} K_{\mathfrak{p}}^{*n} \times \prod_{\mathfrak{p}\in T} K_{\mathfrak{p}}^* \times \prod_{\mathfrak{p}\notin S\cup T} U_{\mathfrak{p}}.$$

Then one has

$$N_{L|K}C_L \supseteq C_K(S,T) \quad \textit{and} \quad \big(C_K : C_K(S,T)\big) = [L:K].$$

In particular, if $L|K$ is cyclic, then $N_{L|K}C_L = C_K(S,T)$.

Remark: It will follow from (5.5) that $N_{L|K}C_L = C_K(S,T)$ also holds in general.

For the proof of the theorem we need the following

(4.3) Lemma. $I_K(S,T) \cap K^* = (K^{S\cup T})^n$.

Proof: The inclusion $(K^{S\cup T})^n \subseteq I_K(S,T) \cap K^*$ is trivial. Let $y \in I_K(S,T) \cap K^*$, and $M = K(\sqrt[n]{y})$. It suffices to show that $N_{M|K}C_M = C_K$, for then (3.6) implies $M = K$, hence $y \in K^{*n} \cap I_K(S,T) \subseteq (K^{S\cup T})^n$. Let $[\alpha] \in C_K = I_K^S K^*/K^*$, and let $\alpha \in I_K^S$ be a representative of the class $[\alpha]$. The map

$$K^S \longrightarrow \prod_{\mathfrak{p}\in T} U_{\mathfrak{p}}/U_{\mathfrak{p}}^n$$

is surjective. For if Δ denotes its kernel, then obviously $K^{*n} \cap \Delta = (K^S)^n$, and $\Delta K^{*n}/K^{*n} = \Delta/(K^S)^n$. From (1.1) and Kummer theory, we therefore get

$$\#(K^S/\Delta) = \frac{\#K^S/(K^S)^n}{\#\Delta/(K^S)^n} = \frac{n^s}{\#G(L|K)} = n^{s-r}.$$

This is also the order of the product because by chap. II, (5.8), we have $\#U_{\mathfrak{p}}/U_{\mathfrak{p}}^n = n$ since $\mathfrak{p} \nmid n$. We thus find an element $x \in K^S$ such that $\alpha_{\mathfrak{p}} = xu_{\mathfrak{p}}^n$, $u_{\mathfrak{p}} \in U_{\mathfrak{p}}$, for $\mathfrak{p} \in T$. The idèle $\alpha' = \alpha x^{-1}$ belongs to the same class as α, and we show that $\alpha' \in N_{M|K}I_M$. By (3.2), this amounts to checking that every component $\alpha'_{\mathfrak{p}}$ is a norm from $M_{\mathfrak{P}}|K_{\mathfrak{p}}$. For $\mathfrak{p} \in S$ this holds because $y \in K_{\mathfrak{p}}^{*n}$. Hence we have $M_{\mathfrak{P}} = K_{\mathfrak{p}}$ for $\mathfrak{p} \in T$ since $\alpha'_{\mathfrak{p}} = u_{\mathfrak{p}}^n$ is a n-th power. For $\mathfrak{p} \notin S \cup T$ it holds because $\alpha'_{\mathfrak{p}}$ is a unit and $M_{\mathfrak{P}}|K_{\mathfrak{p}}$ is unramified (see chap. V, (3.3)). This is why $[\alpha] \in N_{M|K}C_M$, q.e.d. □

Proof of theorem (4.2): The identity $(C_K : C_K(S,T)) = [L : K]$ follows from the exact sequence

$$\begin{aligned}1 \longrightarrow I_K^{S\cup T} \cap K^*/I_K(S,T) \cap K^* \longrightarrow I_K^{S\cup T}/I_K(S,T)& \\ \longrightarrow I_K^{S\cup T}K^*/I_K(S,T)K^* \longrightarrow 1&.\end{aligned}$$

Since $I_K^{S\cup T}K^* = I_K$, the order of the group on the right is

$$\begin{aligned}\big(I_K^{S\cup T}K^* : I_K(S,T)K^*\big) &= \big(I_KK^*/K^* : I_K(S,T)K^*/K^*\big)\\ &= \big(C_K : C_K(S,T)\big).\end{aligned}$$

The order of the group on the left is

$$\big(I_K^{S\cup T} \cap K^* : I_K(S,T) \cap K^*\big) = \big(K^{S\cup T} : (K^{S\cup T})^n\big) = n^{2s-r}$$

because $\#(S \cup T) = 2s - r$, and $\mu_n \subseteq K^{S\cup T}$. In view of chap. II, (5.8), the order of the group in the middle is

$$\big(I_K^{S\cup T} : I_K(S,T)\big) = \prod_{\mathfrak{p}\in S}(K_{\mathfrak{p}}^* : K_{\mathfrak{p}}^{*n}) = \prod_{\mathfrak{p}\in S}\frac{n^2}{|n|_{\mathfrak{p}}} = n^{2s}\prod_{\mathfrak{p}}|n|_{\mathfrak{p}}^{-1} = n^{2s}.$$

Altogether this gives

$$\big(C_K : C_K(S,T)\big) = \frac{n^{2s}}{n^{2s-r}} = n^r = [L:K].$$

We now show the inclusion $C_K(S,T) \subseteq N_{L|K}C_L$. Let $\alpha \in I_K(S,T)$. In order to show that $\alpha \in N_{L|K}I_L$ all we have to check, by (3.2), is again that every component $\alpha_{\mathfrak p}$ is a norm from $L_{\mathfrak P}|K_{\mathfrak p}$. For $\mathfrak p \in S$ this is true because $\alpha_{\mathfrak p} \in K_{\mathfrak p}^{*n}$ is an n-th power, hence a norm from $K_{\mathfrak p}(\sqrt[n]{K_{\mathfrak p}^*})$ (see chap. V, (1.5)), so in particular also from $L_{\mathfrak P}|K_{\mathfrak p}$. For $\mathfrak p \in T$ it holds because (4.1) gives $\Delta \subseteq K_{\mathfrak p}^{*n}$, and thus $L_{\mathfrak P} = K_{\mathfrak p}$. Finally, it holds for $\mathfrak p \notin S \cup T$ since $\alpha_{\mathfrak p}$ is a unit and $L_{\mathfrak P}|K_{\mathfrak p}$ is unramified (see chap. V, (3.3)). We therefore have $I_K(S,T) \subseteq N_{L|K}I_L$, i.e., $C_K(S,T) \subseteq N_{L|K}C_L$.

Now if $L|K$ is cyclic, i.e., if $r = 1$, then from (3.6),

$$[L:K] \le (C_K : N_{L|K}C_L) \le \big(C_K : C_K(S,T)\big) = [L:K],$$

hence $N_{L|K}C_L = C_K(S,T)$. □

Now that we have an explicit picture in the case of a Kummer field, the result we want follows also in complete generality:

(4.4) Theorem (Global Class Field Axiom). *If $L|K$ is a cyclic extension of algebraic number fields, then*

$$\#H^i\big(G(L|K), C_L\big) = \begin{cases} [L:K] & \text{for } i = 0, \\ 1 & \text{for } i = -1. \end{cases}$$

Proof: Since $h(G(L|K), C_L) = [L:K]$, it is clearly enough to show that $\#H^0(G(L|K), C_L) \mid [L:K]$. We will prove this by induction on the degree $n = [L:K]$. We write for short $H^0(L|K)$ instead of $H^0(G(L|K), C_L)$. Let $M|K$ be a subextension of prime degree p. We consider the exact sequence

$$C_M/N_{L|M}C_L \xrightarrow{N_{M|K}} C_K/N_{L|K}C_L \longrightarrow C_K/N_{M|K}C_M \longrightarrow 1,$$

i.e., the exact sequence

$$H^0(L|M) \longrightarrow H^0(L|K) \longrightarrow H^0(M|K) \longrightarrow 1.$$

If $p < n$, then $\#H^0(L|M) \mid [L:M]$, $\#H^0(M|K) \mid [M:K]$ by the induction hypothesis, hence $\#H^0(L|K) \mid [L:M][M:K] = [L:K]$.

Now let $p = n$. We put $K' = K(\mu_p)$ and $L' = L(\mu_p)$. Since $d = [K':K] \mid p-1$, we have $G(L|K) \cong G(L'|K')$. $L'|K'$ is a cyclic

Kummer extension, so by (4.2), $\#H^0(L'|K') = [L' : K'] = p$. It therefore suffices to show that the homomorphism

$$(*) \qquad H^0(L|K) \longrightarrow H^0(L'|K')$$

induced by the inclusion $C_L \to C_{L'}$ is injective. $H^0(L|K)$ has exponent p, because for $x \in C_K$ we always have $x^p = N_{L|K}(x)$. Taking $d = [K' : K]$-th powers on $H^0(L|K)$ is therefore an isomorphism. Now let $\bar{x} = x \bmod N_{L|K}C_L$ belong to the kernel of $(*)$. We write $\bar{x} = \bar{y}^d$, for some $\bar{y} = y \bmod N_{L|K}C_L$. Then $\bar{y}$ also is in the kernel of $(*)$, i.e., $y = N_{L'|K'}(z')$, $z' \in C_{L'}$, and we find:

$$y^d = N_{K'|K}(y) = N_{L'|K}(z') = N_{L|K}(N_{L'|L}(z')) \in N_{L|K}C_L\,.$$

Hence $\bar{x} = \bar{y}^d = 1$. □

An immediate consequence of the theorem we have just proved is the famous **Hasse Norm Theorem**:

(4.5) Corollary. *Let $L|K$ be a cyclic extension. An element $x \in K^*$ is a norm if and only if it is a norm locally everywhere, i.e., a norm in every completion $L_{\mathfrak{P}}|K_{\mathfrak{p}}$ $(\mathfrak{P}|\mathfrak{p})$.*

Proof: Let $G = G(L|K)$ and $G_{\mathfrak{P}} = G(L_{\mathfrak{P}}|K_{\mathfrak{p}})$. The exact sequence

$$1 \longrightarrow L^* \longrightarrow I_L \longrightarrow C_L \longrightarrow 1$$

of G-modules gives, by chap. IV, (7.1), an exact sequence

$$H^{-1}(G, C_L) \longrightarrow H^0(G, L^*) \longrightarrow H^0(G, I_L)\,.$$

By (4.4), we have $H^{-1}(G, C_L) = 1$, and from (3.2) it follows that $H^0(G, I_L) = \bigoplus_{\mathfrak{p}} H^0(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*)$. Therefore the homomorphism

$$K^*/N_{L|K}L^* \longrightarrow \bigoplus_{\mathfrak{p}} K_{\mathfrak{p}}^*/N_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}L_{\mathfrak{P}}^*$$

is injective. But this is the claim of the corollary. □

It should be noted that cyclicity is crucial for Hasse's norm theorem. In fact, whereas it is true by (3.2) that an element $x \in K^*$ which is everywhere locally a norm, is always the norm of some idèle α of L, this need not be by any means a principal idèle, not even in the case of arbitrary abelian extensions.

Exercise 1. Determine the norm group $N_{L|K}C_L$ for an arbitrary Kummer extension in a way analogous to the case treated in (4.2) where $G(L|K) \cong (\mathbb{Z}/p^a\mathbb{Z})^r$.

Exercise 2. Let ζ be a primitive m-th root of unity. Show that the norm group $N_{\mathbb{Q}(\zeta)|\mathbb{Q}}C_{\mathbb{Q}}$ equals the ray class group mod $\mathfrak{m} = (m)$ in $C_{\mathbb{Q}}$.

Exercise 3. An equation $x^2 - ay^2 = b$, $a, b \in K^*$, has a solution in K if and only if it is solvable everywhere locally, i.e., in each completion $K_{\mathfrak{p}}$.

Hint: $x^2 - ay^2 = N_{K(\sqrt{a})|K}(x - \sqrt{a}y)$ if $a \notin K^{*2}$.

Exercise 4. If a quadratic form $a_1x_1^2 + \cdots + a_nx_n^2$ represents zero over a field K with more than five elements (i.e., $a_1x_1^2 + \cdots + a_nx_n^2 = 0$ has a nontrivial solution in K), then there is a representation of zero in which all $x_i \neq 0$.

Hint: If $a\xi^2 = \lambda \neq 0$, $b \neq 0$, then there are non-zero elements α and β such that $a\alpha^2 + b\beta^2 = \lambda$. To prove this, multiply the identity

$$\frac{(t-1)^2}{(t+1)^2} + \frac{4t}{(t+1)^2} = 1$$

by $a\xi^2 = \lambda$ and insert $t = b\gamma^2/a$, for some element $\gamma \neq 0$ such that $t \neq \pm 1$. Use this to prove the claim by induction.

Exercise 5. A quadratic form $ax^2 + by^2 + cz^2$, $a, b, c \in K^*$, represents zero if and only if it represents zero everywhere locally.

Remark: In complete generality, one has the following "local-to-global principle":

Theorem of Minkowski-Hasse: A quadratic form over a number field K represents zero if and only if it represents zero over every completion $K_{\mathfrak{p}}$.

The proof follows from the result stated in exercise 5 by pure algebra (see [113]).

§ 5. The Global Reciprocity Law

Now that we know that the idèle class group satisfies the class field axiom, we proceed to determine a pair of homomorphisms

$$(G_{\mathbb{Q}} \xrightarrow{d} \widehat{\mathbb{Z}},\ C_{\mathbb{Q}} \xrightarrow{v} \widehat{\mathbb{Z}})$$

obeying the rules of abstract class field theory as developed in chap. IV, §4. For the $\widehat{\mathbb{Z}}$-extension of $\mathbb{Q}$ given by d, we have only one choice. It is described in the following

(5.1) Proposition. *Let $\Omega|\mathbb{Q}$ be the field obtained by adjoining all roots of unity, and let T be the torsion subgroup of $G(\Omega|K)$ (i.e., the group of all elements of finite order). Then the fixed field $\widetilde{\mathbb{Q}}|\mathbb{Q}$ of T is a $\widehat{\mathbb{Z}}$-extension.*

Proof: Since $\Omega = \bigcup_{n\geq 1}\mathbb{Q}(\mu_n)$, we find

$$G(\Omega|\mathbb{Q}) = \varprojlim_n G\big(\mathbb{Q}(\mu_n)|\mathbb{Q}\big) = \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^* = \widehat{\mathbb{Z}}^*.$$

But $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, and $\mathbb{Z}_p^* \cong \mathbb{Z}_p \times \mathbb{Z}/(p-1)\mathbb{Z}$ for $p \neq 2$ and $\mathbb{Z}_2^* \cong \mathbb{Z}_2 \times \mathbb{Z}/2\mathbb{Z}$. Consequently,

$$G(\Omega|\mathbb{Q}) \cong \widehat{\mathbb{Z}}^* \cong \widehat{\mathbb{Z}} \times \widehat{T}, \quad \text{where} \quad \widehat{T} = \prod_{p\neq 2} \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

This shows that the torsion subgroup T of $G(\Omega|\mathbb{Q})$ is isomorphic to the torsion subgroup of $\widehat{T}$. Since the latter contains the group $\bigoplus_{p\neq 2}\mathbb{Z}/(p-1)\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, we see that the closure $\overline{T}$ of T is isomorphic to $\widehat{T}$. Now, if $\widetilde{\mathbb{Q}}$ is the fixed field of T, this implies that $G(\widetilde{\mathbb{Q}}|\mathbb{Q}) = G(\Omega|\mathbb{Q})/\overline{T} \cong \widehat{\mathbb{Z}}$. □

Another description of the $\widehat{\mathbb{Z}}$-extension $\widetilde{\mathbb{Q}}|\mathbb{Q}$ is obtained in the following manner. For every prime number p, let $\Omega_p|\mathbb{Q}$ be the field obtained by adjoining all roots of unity of p-power order. Then

$$G(\Omega_p|\mathbb{Q}) = \varprojlim_\nu G\big(\mathbb{Q}(\mu_{p^\nu})|\mathbb{Q}\big) = \varprojlim_\nu (\mathbb{Z}/p^\nu\mathbb{Z})^* = \mathbb{Z}_p^*,$$

and $\mathbb{Z}_p^* \cong \mathbb{Z}_p \times \mathbb{Z}/(p-1)\mathbb{Z}$ for $p \neq 2$ and $\mathbb{Z}_2^* \cong \mathbb{Z}_2 \times \mathbb{Z}/2\mathbb{Z}$. The torsion subgroup of $\mathbb{Z}_p^*$ is isomorphic to $\mathbb{Z}/(p-1)\mathbb{Z}$, resp. $\mathbb{Z}/2\mathbb{Z}$, and taking its fixed field gives an extension $\widetilde{\mathbb{Q}}^{(p)}|\mathbb{Q}$ with Galois group

$$G(\widetilde{\mathbb{Q}}^{(p)}|\mathbb{Q}) \cong \mathbb{Z}_p.$$

The $\widehat{\mathbb{Z}}$-extension $\widetilde{\mathbb{Q}}|\mathbb{Q}$ is then the composite $\widetilde{\mathbb{Q}} = \prod_p \widetilde{\mathbb{Q}}^{(p)}$.

We fix an isomorphism $G(\widetilde{\mathbb{Q}}|\mathbb{Q}) \cong \widehat{\mathbb{Z}}$. There is no canonical choice as in the case of local fields. However, the reciprocity law will not depend on the choice. Via $G(\widetilde{\mathbb{Q}}|\mathbb{Q}) \cong \widehat{\mathbb{Z}}$, we obtain a continuous surjective homomorphism

$$d : G_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}$$

of the absolute Galois group $G_{\mathbb{Q}} = G(\overline{\mathbb{Q}}|\mathbb{Q})$. With this we continue as in chap. IV, § 4, choosing $k = \mathbb{Q}$ as our base field. If $K|\mathbb{Q}$ is a finite extension, then we put $f_K = [K \cap \widetilde{\mathbb{Q}} : \mathbb{Q}]$ and get a surjective homomorphism

$$d_K = \frac{1}{f_K} d : G_K \longrightarrow \widehat{\mathbb{Z}},$$

which defines the $\widehat{\mathbb{Z}}$-extension $\widetilde{K} = K\widetilde{\mathbb{Q}}$ of K. $\widetilde{K}|K$ is called the **cyclotomic $\widehat{\mathbb{Z}}$-extension** of K. We denote again by φ_K the element of $G(\widetilde{K}|K)$ which is mapped to 1 by the isomorphism $G(\widetilde{K}|K) \cong \widehat{\mathbb{Z}}$, and by $\varphi_{L|K}$ the restriction $\varphi_K|_L$ if $L|K$ is a subextension of $\widetilde{K}|K$. The automorphism $\varphi_{L|K}$ must not be confused with the Frobenius automorphism corresponding to a prime ideal of L (see §7).

For the $G_{\mathbb{Q}}$-module A, we choose the union of the idèle class groups C_K of all finite extensions $K|\mathbb{Q}$. Thus $A_K = C_K$. The henselian valuation $v : C_{\mathbb{Q}} \to \widehat{\mathbb{Z}}$ will be obtained as the composite

$$C_{\mathbb{Q}} \xrightarrow{[\;,\widetilde{\mathbb{Q}}|\mathbb{Q}]} G(\widetilde{\mathbb{Q}}|\mathbb{Q}) \xrightarrow{d} \widehat{\mathbb{Z}},$$

where the mapping $[\;,\widetilde{\mathbb{Q}}|\mathbb{Q}]$ will later turn out to be the norm residue symbol $(\;,\widetilde{\mathbb{Q}}|\mathbb{Q})$ of global class field theory (see (5.7)). For the moment we merely define it as follows.

For an arbitrary finite abelian extension $L|K$, we define the homomorphism

$$[\;,L|K] : I_K \longrightarrow G(L|K)$$

by

$$[\alpha, L|K] = \prod_{\mathfrak{p}} (\alpha_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}}),$$

where $L_{\mathfrak{p}}$ denotes the completion of L with respect to a place $\mathfrak{P}|\mathfrak{p}$, and $(\alpha_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}})$ is the norm residue symbol of local class field theory. Note that almost all factors in the product are 1 because almost all extensions $L_{\mathfrak{p}}|K_{\mathfrak{p}}$ are unramified and almost all $\alpha_{\mathfrak{p}}$ are units.

(5.2) Proposition. *If $L|K$ and $L'|K'$ are two abelian extensions of finite algebraic number fields such that $K \subseteq K'$ and $L \subseteq L'$, then we have the commutative diagram*

$$\begin{tikzcd}
I_{K'} \arrow[r, "{[\;,L'|K']}"] \arrow[d, "N_{K'|K}"'] & G(L'|K') \arrow[d] \\
I_K \arrow[r, "{[\;,L|K]}"] & G(L|K).
\end{tikzcd}$$

Proof: For an idèle $\alpha = (\alpha_{\mathfrak{P}}) \in I_{K'}$ of K', we find by chap. IV, (6.4), that

$$(\alpha_{\mathfrak{P}}, L'_{\mathfrak{P}}\,|\,K'_{\mathfrak{P}})|_{L_{\mathfrak{p}}} = \big(N_{K'_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{P}}), L_{\mathfrak{p}}|K_{\mathfrak{p}}\big)\,, \quad (\mathfrak{P}|\mathfrak{p}),$$

and (2.2) implies

$$\begin{aligned}[N_{K'|K}(\alpha), L|K] &= \prod_{\mathfrak{p}}(N_{K'|K}(\alpha)_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}}) = \prod_{\mathfrak{p}}\prod_{\mathfrak{P}|\mathfrak{p}}(N_{K'_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha_{\mathfrak{P}}), L_{\mathfrak{p}}|K_{\mathfrak{p}}) \\ &= \prod_{\mathfrak{P}}(\alpha_{\mathfrak{P}}, L'_{\mathfrak{P}}|K'_{\mathfrak{P}})|_L = [\alpha, L'|K']|_L . \qquad \square\end{aligned}$$

If $L|K$ is an abelian extension of infinite degree, then we define the homomorphism

$$[\ \ ,L|K] : I_K \longrightarrow G(L|K)$$

by its restrictions $[\ \ ,L|K]|_{L'} := [\ \ ,L'|K]$ to the finite subextensions L' of $L|K$. In other words, if $\alpha \in I_K$, then the elements $[\alpha, L'|K]$ define, by (5.2), an element of the projective limit $\varprojlim_{L'} G(L'|K)$, and $[\alpha, L|K]$ is precisely this element, once we identify $G(L|K) = \varprojlim G(L'|K)$. Again one has the equation

$$[\alpha, L|K] = \prod_{\mathfrak{p}}(\alpha_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}}),$$

where $L_{\mathfrak{p}}$ does not denote the completion of L with respect to a place above $\mathfrak{p}$, but rather the *localization*, i.e., the union of the completions $L'_{\mathfrak{p}}|K_{\mathfrak{p}}$ of all finite subextensions (see chap. II, §8). Then $L_{\mathfrak{p}}|K_{\mathfrak{p}}$ is Galois, $G(L_{\mathfrak{p}}|K_{\mathfrak{p}}) \subseteq G(L|K)$, and the product $\prod_{\mathfrak{p}}(\alpha_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}})$ converges in the profinite group to the element $[\alpha, L|K]$. Indeed, if $L'|K$ varies over the finite subextensions of $L|K$, then the sets $S_{L'} = \{\mathfrak{p} \mid (\alpha_{\mathfrak{p}}, L'_{\mathfrak{p}}|K_{\mathfrak{p}}) \neq 1\}$ are all finite, so that we may write down the finite products

$$\sigma_{L'} = \prod_{\mathfrak{p}\in S_{L'}}(\alpha_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}}) \in G(L|K).$$

They converge to $[\alpha, L|K]$, for if $[\alpha, L|K]G(L|N)$ is one of the fundamental neighbourhoods (i.e., $N|K$ is one of the finite subextensions of $L|K$), then

$$\sigma_{L'} \in [\alpha, L|K]G(L|N)$$

for all $L' \supseteq N$ because

$$\sigma_{L'}|_N = \prod_{\mathfrak{p}}(\alpha_{\mathfrak{p}}, N_{\mathfrak{p}}|K_{\mathfrak{p}}) = [\alpha, N|K] = [\alpha, L|K]|_N .$$

This shows that $[\alpha, L|K]$ is the only accumulation point of the family $\{\sigma_{L'}\}$.

It is clear that proposition (5.2) remains true for infinite extensions L and L' of finite algebraic number fields K and K'.

(5.3) Proposition. *For every root of unity ζ and every principal idèle $a \in K^*$ one has*

$$[a,\ K(\zeta)|K] = 1.$$

Proof: By (5.2), we have $[N_{K|\mathbb{Q}}(a), \mathbb{Q}(\zeta)|\mathbb{Q}] = [a, K(\zeta)|K]|_{\mathbb{Q}(\zeta)}$. Hence we may assume that $K = \mathbb{Q}$. Likewise we may assume that ζ has prime power order $\ell^m \neq 2$. Now let $a \in \mathbb{Q}^*$, let v_p be the normalized exponential valuation of $\mathbb{Q}$ for $p \neq \infty$ and write $a = u_p p^{v_p(a)}$. For $p \neq \ell, \infty$, $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ is unramified and $(p, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p)$ is the Frobenius automorphism $\varphi_p : \zeta \to \zeta^p$. From chap. V, (2.4), we thus get

$$\big(a, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big)\,\zeta = \zeta^{n_p} \quad \text{with} \quad n_p = \begin{cases} p^{v_p(a)} & \text{for } p \neq \ell, \infty, \\ u_p^{-1} & \text{for } p = \ell, \\ \operatorname{sgn}(a) & \text{for } p = \infty. \end{cases}$$

Hence

$$[a, \mathbb{Q}(\zeta)|\mathbb{Q}]\zeta = \prod_p \big(a, \mathbb{Q}_p(\zeta)|\mathbb{Q}_p\big)\,\zeta = \zeta^\alpha$$

where $\alpha = \prod\limits_p n_p = \operatorname{sgn}(a) \prod\limits_{p \neq \ell, \infty} p^{v_p(a)} u_\ell^{-1} = \operatorname{sgn}(a) \prod\limits_{p \neq \infty} p^{v_p(a)} a^{-1} = 1.$ □

Since the extension $\widetilde{K}|K$ is contained in the field of all roots of unity over K, the proposition implies

$$[a, \widetilde{K}|K] = 1$$

for all $a \in K^*$. The homomorphism $[\ \ , \widetilde{K}|K] : I_K \to G(\widetilde{K}|K)$ therefore induces a homomorphism

$$[\ \ , \widetilde{K}|K] : C_K \longrightarrow G(\widetilde{K}|K),$$

and we consider its composite

$$v_K : C_K \longrightarrow \widehat{\mathbb{Z}}$$

with $d_K : G(\widetilde{K}|K) \to \widehat{\mathbb{Z}}$. The pair (d_K, v_K) is then a class field theory, for we have the

(5.4) Proposition. *The map $v_K : C_K \to \widehat{\mathbb{Z}}$ is surjective and is a henselian valuation with respect to d_K.*

Proof: We first show surjectivity. If $L|K$ is a finite subextension of $\widetilde{K}|K$, then the map

$$[\ ,L|K] = \prod_{\mathfrak{p}}(\ ,L_{\mathfrak{p}}|K_{\mathfrak{p}}) : I_K \longrightarrow G(L|K)$$

is surjective. Indeed, since $(\ ,L_{\mathfrak{p}}|K_{\mathfrak{p}}) : K_{\mathfrak{p}}^* \to G(L_{\mathfrak{p}}|K_{\mathfrak{p}})$ is surjective, $[I_K,L|K]$ contains all decomposition groups $G(L_{\mathfrak{p}}|K_{\mathfrak{p}})$. Thus all $\mathfrak{p}$ split completely in the fixed field M of $[I_K,L|K]$. By (3.8), this implies that $M = K$, and so $[I_K,L|K] = G(L|K)$. This yields furthermore that $[I_K,\widetilde{K}|K] = [C_K,\widetilde{K}|K]$ is dense in $G(\widetilde{K}|K)$. In the exact sequence

$$1 \longrightarrow C_K^0 \longrightarrow C_K \xrightarrow{\mathfrak{N}} \mathbb{R}_+^* \longrightarrow 1$$

(see § 1) the group C_K^0 is compact by (1.6), and we obtain a splitting, if we identify $\mathbb{R}_+^*$ with the group of positive real numbers in any infinite completion $K_{\mathfrak{p}}$. Thus $C_K = C_K^0 \times \mathbb{R}_+^*$. Now, $[\mathbb{R}_+^*,\widetilde{K}|K] = 1$, for if $x \in \mathbb{R}_+^*$, then $[x,\widetilde{K}|K]\,|_L = [x,L|K] = 1$ for every finite subextension $L|K$ of $\widetilde{K}|K$, because we may always write $x = y^n$ with $y \in \mathbb{R}_+^*$ and $n = [L:K]$. Therefore $[C_K,\widetilde{K}|K] = [C_K^0,\widetilde{K}|K]$ is a closed, dense subgroup of $G(\widetilde{K}|K)$ and therefore equal to $G(\widetilde{K}|K)$. This proves the surjectivity of $v_K = d_K \circ [\ ,\widetilde{K}|K]$.

In the definition of a henselian valuation given in chap. IV, (4.6), condition (i) is satisfied because $v_K(C_K) = \widehat{\mathbb{Z}}$, and condition (ii) follows from (5.2) because for every finite extension $L|K$ we have the identity

$$\begin{aligned} v_K(N_{L|K}C_L) &= v_K(N_{L|K}I_L) = d_K[N_{L|K}I_L,\widetilde{K}|K] \\ &= f_{L|K}d_L[I_L,\widetilde{L}|L] = f_{L|K}v_L(C_L) = f_{L|K}\widehat{\mathbb{Z}}. \end{aligned}$$

□

In view of the fact that the idèle class group C_K satisfies the class field axiom, the pair

$$(d_{\mathbb{Q}} : G_{\mathbb{Q}} \to \widehat{\mathbb{Z}},\ v_{\mathbb{Q}} : C_{\mathbb{Q}} \to \widehat{\mathbb{Z}})$$

constitutes a class field theory, the "global class field theory". The above homomorphism $v_K = d_K \circ [\ \ ,\widetilde{K}|K] : C_K \to \widehat{\mathbb{Z}}$, for finite extensions $K|\mathbb{Q}$, satisfies the formula

$$v_K = \frac{1}{f_K} d_{\mathbb{Q}} \circ [\ \ ,\widetilde{\mathbb{Q}}|\mathbb{Q}] \circ N_{K|\mathbb{Q}} = \frac{1}{f_K} v_{\mathbb{Q}} \circ N_{K|\mathbb{Q}}$$

and is therefore precisely the induced homomorphism in the sense of the abstract theory in chap. IV, (4.7).

As the main result of global class field theory we now obtain the **Artin reciprocity law:**

(5.5) Theorem. *For every Galois extension $L|K$ of finite algebraic number fields we have a canonical isomorphism*

$$r_{L|K} : G(L|K)^{ab} \xrightarrow{\sim} C_K/N_{L|K}C_L\,.$$

The inverse map of $r_{L|K}$ yields a surjective homomorphism

$$(\ \ ,L|K) : C_K \to G(L|K)^{ab}$$

with kernel $N_{L|K}C_L$. The map $(\ ,L|K)$ is called the **global norm residue symbol**. We view it also as a homomorphism $I_K \to G(L|K)^{ab}$.

For every place $\mathfrak{p}$ of K, we have on the one hand the embedding $G(L_\mathfrak{p}|K_\mathfrak{p}) \hookrightarrow G(L|K)$, and on the other the canonical injection

$$\langle\ \ \rangle : K_\mathfrak{p}^* \to C_K\,,$$

which sends $a_\mathfrak{p} \in K_\mathfrak{p}^*$ to the class of the idèle

$$\langle a_\mathfrak{p}\rangle = (\dots, 1, 1, 1, a_\mathfrak{p}, 1, 1, 1, \dots).$$

These homomorphisms express the compatibility of local and global class field theory, as follows.

(5.6) Proposition. *If $L|K$ is an abelian extension and $\mathfrak{p}$ is a place of K, then the diagram*

$$\begin{tikzcd} K_\mathfrak{p}^* \arrow[r, "(\ ,L_\mathfrak{p}|K_\mathfrak{p})"] \arrow[d, "\langle\ \rangle"'] & G(L_\mathfrak{p}|K_\mathfrak{p}) \arrow[d] \\ C_K \arrow[r, "(\ ,L|K)"] & G(L|K) \end{tikzcd}$$

is commutative.

Proof: We first show that the proposition holds if $L|K$ is a subextension of $\widetilde{K}|K$, or if $L = K(i)$, $i = \sqrt{-1}$, and $\mathfrak{p}|\infty$. Indeed, the two maps $[\ ,\widetilde{K}|K]$, $(\ ,\widetilde{K}|K) : I_K \to G(\widetilde{K}|K)$ agree because from chap. IV, (6.5), we have

$$d_K \circ (\ ,\widetilde{K}|K) = v_K = d_K \circ [\ ,\widetilde{K}|K]\,.$$

Thus, if $L|K$ is a subextension of $\widetilde{K}|K$ and $\alpha = (\alpha_\mathfrak{p}) \in I_K$, then

$$(\alpha, L|K) = [\alpha, L|K] = \prod_\mathfrak{p} (\alpha_\mathfrak{p}, L_\mathfrak{p}|K_\mathfrak{p})\,.$$

In particular, for $a_\mathfrak{p} \in K_\mathfrak{p}^*$ we have the identity

$$(\langle a_\mathfrak{p}\rangle, L|K) = (a_\mathfrak{p}, L_\mathfrak{p}|K_\mathfrak{p})\,,$$

which shows that the diagram is commutative when restricted to the finite subextensions of $\widetilde{K}|K$.

On the other hand, let $L = K(i)$, $\mathfrak{p}|\infty$, and $L_\mathfrak{p} \neq K_\mathfrak{p}$. Then $K_\mathfrak{p}^* = \mathbb{R}^*$, $\mathbb{R}_+^*$ is the kernel of $(\ ,L_\mathfrak{p}|K_\mathfrak{p})$, and $(-1, L_\mathfrak{p}|K_\mathfrak{p})$ is complex conjugation in $G(L_\mathfrak{p}|K_\mathfrak{p}) = G(\mathbb{C}|\mathbb{R})$. Thus, all we have to show is that $(\langle -1\rangle, L|K) \neq 1$. If we had $(\langle -1\rangle, L|K) = 1$, then the class of $\langle -1\rangle$ would be the norm of a class of C_L, i.e., $\langle -1\rangle a = N_{L|K}(\alpha)$ for some $a \in K^*$ and an idèle $\alpha \in I_L$. This would mean that $a = N_{L_\mathfrak{q}|K_\mathfrak{q}}(\alpha_\mathfrak{q})$ for $\mathfrak{q} \neq \mathfrak{p}$ and $-a = N_{L_\mathfrak{p}|K_\mathfrak{p}}(\alpha_\mathfrak{p})$, i.e., $(a, L_\mathfrak{q}|K_\mathfrak{q}) = 1$ for $\mathfrak{q} \neq \mathfrak{p}$ and $(-a, L_\mathfrak{p}|K_\mathfrak{p}) = 1$. By (5.3), we would have $1 = [a, L|K] = \prod_\mathfrak{q}(a, L_\mathfrak{q}|K_\mathfrak{q}) = (a, L_\mathfrak{p}|K_\mathfrak{p})$, so that $(-1, L_\mathfrak{p}|K_\mathfrak{p}) = 1$, and therefore $-1 \in N_{L_\mathfrak{p}|K_\mathfrak{p}}(L_\mathfrak{p}^*) = N_{\mathbb{C}|\mathbb{R}}\mathbb{C}^* = \mathbb{R}_+^*$, a contradiction.

We now reduce the general case to these special cases as follows. Let $L'|K'$ be an abelian extension, so that $K \subseteq K'$, $L \subseteq L'$. We then consider the diagram

$$\begin{tikzcd}
& G(L'_\mathfrak{p}|K'_\mathfrak{p}) \arrow[rr] \arrow[dl] \arrow[dd] & & K'^*_\mathfrak{p}/N_{L'_\mathfrak{p}|K'_\mathfrak{p}}L'^*_\mathfrak{p} \arrow[dl] \arrow[dd] \\
G(L_\mathfrak{p}|K_\mathfrak{p}) \arrow[rr, crossing over] \arrow[dd] & & K_\mathfrak{p}^*/N_{L_\mathfrak{p}|K_\mathfrak{p}}L_\mathfrak{p}^* & \\
& G(L'|K') \arrow[rr] \arrow[dl] & & C_{K'}/N_{L'|K'}C_{L'} \arrow[dl] \\
G(L|K) \arrow[rr] & & C_K/N_{L|K}C_L \arrow[from=uu, crossing over] &
\end{tikzcd}$$

where $L_\mathfrak{p} = K_\mathfrak{p}L$, $K'_\mathfrak{p} = K_\mathfrak{p}K'$, $L'_\mathfrak{p} = K_\mathfrak{p}L'$. In this diagram, the top and bottom are commutative by chap. IV, (6.4), and the sides are commutative for trivial reasons. If now $L'|K'$ is one of the special extensions for which the proposition is already established, then the back diagram is commutative, and hence also the front one, for all elements of $G(L_\mathfrak{p}|K_\mathfrak{p})$ in the image of $G(L'_\mathfrak{p}|K'_\mathfrak{p}) \to G(L_\mathfrak{p}|K_\mathfrak{p})$. This makes it clear that it is enough to find, for every $\sigma \in G(L_\mathfrak{p}|K_\mathfrak{p})$, some special extension $L'|K'$ such that σ lies in the image of $G(L'_\mathfrak{p}|K'_\mathfrak{p})$. It is even sufficient to do this only for all σ of prime power order, because they generate the group. Passing to the fixed field of σ we may assume moreover that $G(L|K)$ is generated by σ.

When $\mathfrak{p}|\infty$ and $L_\mathfrak{p} \neq K_\mathfrak{p}$, i.e., $K_\mathfrak{p} = \mathbb{R}$, $L_\mathfrak{p} = \mathbb{C}$, we put $L' = L(i) \subseteq \mathbb{C}$, and choose for K' the fixed field of the restriction of complex conjugation to L'. Then $L' = K'(i)$ and $K'_\mathfrak{p} = \mathbb{R}$, $L'_\mathfrak{p} = \mathbb{C}$, so the mapping $G(L'_\mathfrak{p}|K'_\mathfrak{p}) \to G(L_\mathfrak{p}|K_\mathfrak{p})$ is surjective.

When $\mathfrak{p} \nmid \infty$, we find the extension $L'|K'$ as follows. Let σ be of p-power order. We denote by $\widehat{K}|K$, resp. $\widehat{L}|L$, the $\mathbb{Z}_p$-extension contained in $\widetilde{K}|K$, resp. $\widetilde{L}|L$, and consider the field diagram

$$
\begin{tikzcd}[column sep=small, row sep=small]
& L_\mathfrak{p} \arrow[rr, dash] \arrow[dd, dash] & & \widehat{L}_\mathfrak{p} \arrow[dd, dash] \\
L \arrow[ur, dash] \arrow[rr, dash] \arrow[dd, dash] & & \widehat{L} \arrow[ur, dash] \arrow[dd, dash] & \\
& K_\mathfrak{p} \arrow[rr, dash] & & \widehat{K}_\mathfrak{p} \\
K \arrow[ur, dash] \arrow[rr, dash] & & \widehat{K} \arrow[ur, dash] &
\end{tikzcd}
$$

with localizations $\widehat{K}_\mathfrak{p} = K_\mathfrak{p}\widehat{K}$, $\widehat{L}_\mathfrak{p} = L_\mathfrak{p}\widehat{L}$ (all fields are considered to lie in a common bigger field). We may now lift $\sigma \in G(L_\mathfrak{p}|K_\mathfrak{p}) = G(L|K)$ to an automorphism $\hat{\sigma}$ of $\widehat{L}_\mathfrak{p}$ such that

(1) $\hat{\sigma} \in G(\widehat{L}_\mathfrak{p}|K_\mathfrak{p})$,

(2) $\hat{\sigma}|_{\widehat{K}} = \varphi^n_{\widehat{K}|K}$ for some $n \in \mathbb{N}$.

Indeed, since $\widehat{K}_\mathfrak{p} = K_\mathfrak{p}\widehat{K} \neq K_\mathfrak{p}$, the group $G(\widehat{K}_\mathfrak{p}|K_\mathfrak{p}) \neq 1$, and thus is of finite index if viewed as subgroup of $G(\widehat{K}|K) \cong \mathbb{Z}_p$. It is therefore generated by a natural power $\psi = \varphi^k_{\widehat{K}|K}$ of Frobenius $\varphi_{\widehat{K}|K} \in G(\widehat{K}|K)$. As in the proof of chap. IV, (4.4), we may then lift σ to a $\hat{\sigma} \in G(\widehat{L}_\mathfrak{p}|K_\mathfrak{p})$ such that $\hat{\sigma}|_{\widehat{K}_\mathfrak{p}} = \psi^m$, $m \in \mathbb{N}$, so that $\hat{\sigma}|_{\widehat{K}} = \varphi^{km}_{\widehat{K}|K}$.

We now take the fixed field K' of $\hat{\sigma}|_{\widehat{L}}$, and the extension $L' = K'L$. As in chap. IV, (4.5), conditions (ii) and (iii), it then follows that $[K' : K] < \infty$ and $\widehat{K}' = \widehat{L}$. $L'|K'$ is therefore a subextension of $\widehat{K}'|K'$, and σ is the image of $\hat{\sigma}|_{L'_\mathfrak{p}}$ under $G(L'_\mathfrak{p}|K'_\mathfrak{p}) \to G(L_\mathfrak{p}|K_\mathfrak{p})$. This finishes the proof. □

(5.7) Corollary. *If $L|K$ is an abelian extension and $\alpha = (\alpha_\mathfrak{p}) \in I_K$, then*

$$(\alpha, L|K) = \prod_\mathfrak{p} (\alpha_\mathfrak{p}, L_\mathfrak{p}|K_\mathfrak{p}).$$

In particular, for a principal idèle $a \in K^*$ we have the product formula

$$\prod_\mathfrak{p} (a, L_\mathfrak{p}|K_\mathfrak{p}) = 1.$$

Proof: Since I_K is topologically generated by the idèles of the form $\alpha = \langle a_\mathfrak{p} \rangle$, $a_\mathfrak{p} \in K^*_\mathfrak{p}$, it is enough to prove the first formula for these idèles. But this is exactly the statement of (5.6):

$$(\alpha, L|K) = (\langle a_\mathfrak{p} \rangle, L|K) = (a_\mathfrak{p}, L_\mathfrak{p}|K_\mathfrak{p}) = \prod_\mathfrak{q} (\alpha_\mathfrak{q}, L_\mathfrak{q}|K_\mathfrak{q}).$$

The product formula is a consequence of the fact that $(\alpha, L|K)$ depends only on the idèle class $\alpha \bmod K^*$. □

Identifying $K_{\mathfrak{p}}^*$ with its image in C_K under the map $a_{\mathfrak{p}} \mapsto \langle a_{\mathfrak{p}} \rangle$, we obtain the following further corollary, where we use the abbreviations $N = N_{L|K}$ and $N_{\mathfrak{p}} = N_{L_{\mathfrak{p}}|K_{\mathfrak{p}}}$.

(5.8) Corollary. *For every finite abelian extension one has*

$$NC_L \cap K_{\mathfrak{p}}^* = N_{\mathfrak{p}} L_{\mathfrak{p}}^* .$$

Proof: For $x_{\mathfrak{p}} \in N_{\mathfrak{p}} L_{\mathfrak{p}}^*$ we see from (5.6) that $(\langle x_{\mathfrak{p}} \rangle, L|K) = (x_{\mathfrak{p}}, L_{\mathfrak{p}}|K_{\mathfrak{p}}) = 1$. Thus the class of $\langle x_{\mathfrak{p}} \rangle$ is contained in NC_L. Therefore $N_{\mathfrak{p}} L_{\mathfrak{p}}^* \subseteq NC_L$. Conversely, let $\bar{\alpha} \in NC_L \cap K_{\mathfrak{p}}^*$. Then $\bar{\alpha}$ is represented on the one hand by a norm idèle $\alpha = N\beta$, $\beta \in I_L$, and on the other hand by an idèle $\langle x_{\mathfrak{p}} \rangle$, $x_{\mathfrak{p}} \in K_{\mathfrak{p}}^*$. This gives $\langle x_{\mathfrak{p}} \rangle a = N\beta$ with $a \in K^*$. Passing to components shows that a is a norm from $L_{\mathfrak{q}}|K_{\mathfrak{q}}$ for every $\mathfrak{q} \neq \mathfrak{p}$, and the product formula (5.7) shows that a is also a norm from $L_{\mathfrak{p}}|K_{\mathfrak{p}}$. Therefore $x_{\mathfrak{p}} \in N_{\mathfrak{p}} L_{\mathfrak{p}}^*$, and this proves the inclusion $NC_L \cap K^* \subseteq N_{\mathfrak{p}} L_{\mathfrak{p}}^*$. □

Exercise 1. If D_K is the connected component of the unit element of C_K, and if $K^{ab}|K$ is the maximal abelian extension of K, then $C_K/D_K \cong G(K^{ab}|K)$.

Exercise 2. For every place $\mathfrak{p}$ of K one has $K_{\mathfrak{p}}^{ab} = K^{ab} K_{\mathfrak{p}}$.

Hint: Use (5.6) and (5.8).

Exercise 3. Let p be a prime number, and let $M_p|K$ be the maximal abelian p-extension unramified outside of $\{\mathfrak{p}|p\}$. Further, let $H|K$ be the maximal unramified subextension of $M_p|K$ in which the infinite places split completely. Then there is an exact sequence

$$1 \to G(M_p|H) \to G(M_p|K) \to Cl_K(p) \to 1 ,$$

where $Cl_K(p)$ is the p-Sylow subgroup of the ideal class group Cl_K, and there is a canonical isomorphism

$$G(M_p|H) \cong \prod_{\mathfrak{p}|p} U_{\mathfrak{p}}^{(1)} / (\prod_{\mathfrak{p}|p} U_{\mathfrak{p}}^{(1)} \cap \bar{E}) ,$$

where $\bar{E}$ is the closure of the (diagonally embedded) unit group $E = \mathcal{O}_K^*$ in $\prod_{\mathfrak{p}|p} U_{\mathfrak{p}}$.

Exercise 4. The group $\bar{E}(p) := \bar{E} \cap \prod_{\mathfrak{p}|p} U_{\mathfrak{p}}^{(1)}$ is a $\mathbb{Z}_p$-module of rank $r_p(E) := \mathrm{rank}_{\mathbb{Z}_p}(\bar{E}(p)) = [K:\mathbb{Q}] - \mathrm{rank}_{\mathbb{Z}_p} G(M_p|K)$. $r_p(E)$ is called the ***p-adic unit rank***.

Problem: For the p-adic unit rank, one has the famous **Leopoldt conjecture**:

$$r_p(E) = r + s - 1 ,$$

where r, resp. s, is the number of all real, resp. complex, places; in other words,

$$\mathrm{rank}_{\mathbb{Z}_p} G(M_p|K) = s + 1 .$$

The Leopoldt conjecture was proved for abelian number fields $K|\mathbb{Q}$ by the American mathematician Armand Brumer [22]. The general case is still open to date.

§ 6. Global Class Fields

As in local class field theory, the reciprocity law provides also in global class field theory a complete classification of all abelian extensions of a finite algebraic number field K. For this it is necessary to view the idèle class group C_K as a topological group, equipped with its natural topology which the valuations of the various completions $K_{\mathfrak{p}}$ impress upon it (see § 1).

(6.1) Theorem. *The map*

$$L \longmapsto \mathcal{N}_L = N_{L|K} C_L$$

is a $1-1$*-correspondence between the finite abelian extensions* $L|K$ *and the closed subgroups of finite index in* C_K. *Moreover one has:*

$$L_1 \subseteq L_2 \Longleftrightarrow \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}.$$

The field $L|K$ *corresponding to the subgroup* $\mathcal{N}$ *of* C_K *is called the* **class field** *of* $\mathcal{N}$. *It satisfies*

$$G(L|K) \cong C_K/\mathcal{N}.$$

Proof: By chap. IV, (6.7), all we have to show is that the subgroups $\mathcal{N}$ of C_K which are open in the norm topology are precisely the closed subgroups of finite index for the natural topology.

If the subgroup $\mathcal{N}$ is open in the norm topology, then it contains a norm group $N_{L|K}C_L$ and is therefore of finite index, because from (5.5), $(C_K : N_{L|K}C_L) = \#G(L|K)^{ab}$. To show that $\mathcal{N}$ is closed it is enough to show that $N_{L|K}C_L$ is. For this, we choose an infinite place $\mathfrak{p}$ of K and denote by Γ_K the image of the subgroup of positive real numbers in $K_{\mathfrak{p}}$ under the mapping $\langle\ \ \rangle : K_{\mathfrak{p}}^* \to C_K$. Then Γ_K is a group of representatives for the homomorphism $\mathfrak{N} : C_K \to \mathbb{R}_+^*$ with kernel C_K^0 (see § 1), i.e., $C_K = C_K^0 \times \Gamma_K$. By the same token, Γ_K is a group of representatives for the homomorphism $\mathfrak{N} : C_L \to \mathbb{R}_+^*$. We therefore get

$$N_{L|K}C_L = N_{L|K}C_L^0 \times N_{L|K}\Gamma_K = N_{L|K}C_L^0 \times \Gamma_K^n = N_{L|K}C_L^0 \times \Gamma_K.$$

The norm map is continuous, and C_L^0 is compact by (1.6). Hence $N_{L|K}C_L^0$ is closed. Since Γ_K is clearly also closed in C_K, we get that $N_{L|K}C_L$ is closed.

Conversely let $\mathcal{N}$ be a closed subgroup of C_K of finite index. We have to show that $\mathcal{N}$ is open in the norm topology, i.e., contains a norm group $N_{L|K}C_L$. For this we may assume that the index n is a prime power. For if $n = p_1^{\nu_1} \cdots p_r^{\nu_r}$, and $\mathcal{N}_i \subseteq C_K$ is the group containing $\mathcal{N}$ of index $p_i^{\nu_i}$, then $\mathcal{N} = \bigcap_{i=1}^r \mathcal{N}_i$, and if the $\mathcal{N}_i$ are open in the norm topology, then so is $\mathcal{N}$.

Now let J be the preimage of $\mathcal{N}$ with respect to the projection $I_K \to C_K$. Then J is open in I_K because $\mathcal{N}$ is open in C_K (with respect to the natural topology). Therefore J contains a group

$$U_K^S = \prod_{\mathfrak{p}\in S}\{1\} \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}},$$

where S is a sufficiently big finite set of places of K containing the infinite ones and those primes that divide n, such that $I_K = I_K^S K^*$. Since $(I_K : J) = n$, J also contains the group $\prod_{\mathfrak{p}\in S} K_{\mathfrak{p}}^{*n} \times \prod_{\mathfrak{p}\notin S}\{1\}$, and hence the group

$$I_K(S) = \prod_{\mathfrak{p}\in S} K_{\mathfrak{p}}^{*n} \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}}.$$

Thus it is enough to show that $C_K(S) = I_K(S)K^*/K^* \subseteq \mathcal{N}$ contains a norm group. If the n-th roots of unity belong to K, then $C_K(S) = N_{L|K}C_L$ with $L = K(\sqrt[n]{K^S})$, because of the remark following (4.2). If they do not belong to K, then we adjoin them and obtain an extension $K'|K$. Let S' be the set of primes of K' lying above primes in S. If S was chosen sufficiently large, then $I_{K'} = I_{K'}^{S'}K'^*$ and $C_{K'}(S') = N_{L'|K'}C_{L'}$, with $L' = K'(\sqrt[n]{K'^{S'}})$, by the above argument. Using chap. V, (1.5), this gives on the other hand that $N_{K'|K}(I_{K'}(S')) \subseteq I_K(S)$, so that

$$N_{L'|K}C_{L'} = N_{K'|K}(N_{L'|K'}C_{L'}) = N_{K'|K}(C_{K'}(S')) \subseteq C_K(S).$$

This finishes the proof. □

The above theorem is called the "existence theorem" of global class field theory because its main assertion is the existence, for any given closed subgroup $\mathcal{N}$ of finite index in C_K, of an abelian extension $L|K$ such that $N_{L|K}C_L = \mathcal{N}$. This extension L is the class field for $\mathcal{N}$. The existence theorem gives a clear overview of all the abelian extensions of K once we bring in the *congruence subgroups* $C_K^{\mathfrak{m}}$ of C_K corresponding to the **modules** $\mathfrak{m} = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{n_{\mathfrak{p}}}$ (see (1.7)). They are closed of finite index by (1.8), and they prompt the following definition.

(6.2) Definition. *The class field $K^{\mathfrak{m}}|K$ for the congruence subgroup $C_K^{\mathfrak{m}}$ is called the* **ray class field** mod $\mathfrak{m}$.

The Galois group of the ray class field is canonically isomorphic to the ray class group mod $\mathfrak{m}$:

$$G(K^{\mathfrak{m}}|K) \cong C_K/C_K^{\mathfrak{m}}.$$

One has

$$\mathfrak{m}|\mathfrak{m}' \implies K^{\mathfrak{m}} \subseteq K^{\mathfrak{m}'},$$

because clearly $C_K^{\mathfrak{m}} \supseteq C_K^{\mathfrak{m}'}$. Since the closed subgroups of finite index in C_K are by (1.8) precisely those subgroups containing a congruence subgroup $C_K^{\mathfrak{m}}$, we get from (6.1) the

(6.3) Corollary. *Every finite abelian extension $L|K$ is contained in a ray class field $K^{\mathfrak{m}}|K$.*

(6.4) Definition. *Let $L|K$ be a finite abelian extension, and let $\mathcal{N}_L = N_{L|K}C_L$. The* **conductor** *$\mathfrak{f}$ of $L|K$ (or of $\mathcal{N}_L$) is the gcd of all modules $\mathfrak{m}$ such that $L \subseteq K^{\mathfrak{m}}$ (i.e., $C_K^{\mathfrak{m}} \subseteq \mathcal{N}_L$).*

$K^{\mathfrak{f}}|K$ is therefore the smallest ray class field containing $L|K$. But it is not true in general that $\mathfrak{m}$ is the conductor of $K^{\mathfrak{m}}|K$. In chap. V, (1.6), we defined the conductor $\mathfrak{f}_{\mathfrak{p}}$ of a $\mathfrak{p}$-adic extension $L_{\mathfrak{p}}|K_{\mathfrak{p}}$ for a finite place $\mathfrak{p}$, to be the smallest power $\mathfrak{f}_{\mathfrak{p}} = \mathfrak{p}^n$ such that $U_K^{(n)} \subseteq N_{L_{\mathfrak{p}}|K_{\mathfrak{p}}}L_{\mathfrak{p}}^*$. For an infinite place $\mathfrak{p}$ we define $\mathfrak{f}_{\mathfrak{p}} = 1$. Then we view $\mathfrak{f}$ as the replete ideal $\mathfrak{f}\prod_{\mathfrak{p}|\infty}\mathfrak{p}^0$ and obtain the

(6.5) Proposition. *If $\mathfrak{f}$ is the conductor of the abelian extension $L|K$, and $\mathfrak{f}_{\mathfrak{p}}$ is the conductor of the local extension $L_{\mathfrak{p}}|K_{\mathfrak{p}}$, then*

$$\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}.$$

Proof: Let $\mathcal{N} = N_{L|K}C_L$, and let $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$ be a module ($n_{\mathfrak{p}} = 0$ for $\mathfrak{p}|\infty$). One then has

$$C_K^{\mathfrak{m}} \subseteq \mathcal{N} \iff \mathfrak{f}|\mathfrak{m} \quad \text{and} \quad \prod_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}|\mathfrak{m} \iff \mathfrak{f}_{\mathfrak{p}}|\mathfrak{p}^{n_{\mathfrak{p}}} \quad \text{for all } \mathfrak{p}.$$

So to prove $\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}$, we have to show the equivalence

$$C_K^{\mathfrak{m}} \subseteq \mathcal{N} \iff \mathfrak{f}_{\mathfrak{p}}|\mathfrak{p}^{n_{\mathfrak{p}}} \quad \text{for all } \mathfrak{p}.$$

It follows from the identity $\mathcal{N} \cap K_{\mathfrak{p}}^* = N_{\mathfrak{p}}L_{\mathfrak{p}}^*$ (see (5.8)):

$$\begin{aligned} C_K^{\mathfrak{m}} \subseteq \mathcal{N} &\iff (\alpha \in I_K^{\mathfrak{m}} \Rightarrow \bar{\alpha} \in \mathcal{N}) \quad \text{for } \alpha \in I_K \\ &\iff (\alpha_{\mathfrak{p}} \equiv 1 \bmod \mathfrak{p}^{n_{\mathfrak{p}}} \Rightarrow \langle \alpha_{\mathfrak{p}} \rangle \in \mathcal{N} \cap K_{\mathfrak{p}}^* = N_{\mathfrak{p}}L_{\mathfrak{p}}^*) \quad \text{for all } \mathfrak{p} \\ &\iff (\alpha_{\mathfrak{p}} \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} \Rightarrow \alpha_{\mathfrak{p}} \in N_{\mathfrak{p}}L_{\mathfrak{p}}^*) \iff U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} \subseteq N_{\mathfrak{p}}L_{\mathfrak{p}}^* \iff \mathfrak{f}_{\mathfrak{p}}|\mathfrak{p}^{n_{\mathfrak{p}}}. \end{aligned}$$

□

By chap. V, (1.7), the local extension $L_{\mathfrak{P}}|K_{\mathfrak{p}}$, for a finite prime $\mathfrak{p}$, is ramified if and only if its conductor $\mathfrak{f}_{\mathfrak{p}}$ is $\neq 1$. This continues to hold also for an infinite place $\mathfrak{p}$, provided we call the extension $L_{\mathfrak{P}}|K_{\mathfrak{p}}$ *unramified* in this case, as we did in chap. III. Then (6.5) yields the

(6.6) Corollary. *Let $L|K$ be a finite abelian extension and $\mathfrak{f}$ its conductor. Then:*

$$\mathfrak{p} \text{ is ramified in } L \iff \mathfrak{p}|\mathfrak{f}.$$

In the case of the base field $\mathbb{Q}$, the ray class fields are nothing but the familiar cyclotomic fields:

(6.7) Proposition. *Let m be a natural number and $\mathfrak{m} = (m)$. Then the ray class field* mod $\mathfrak{m}$ *of $\mathbb{Q}$ is the field*

$$\mathbb{Q}^{\mathfrak{m}} = \mathbb{Q}(\mu_m)$$

of m-th roots of unity.

Proof: Let $m = \prod_{p \neq p_\infty} p^{n_p}$. Then $I_{\mathbb{Q}}^{\mathfrak{m}} = \prod_{p \neq p_\infty} U_p^{(n_p)} \times \mathbb{R}_+^*$. Let $m = m'p^{n_p}$. Then $U_p^{(n_p)}$ is certainly contained in the norm group of the unramified extension $\mathbb{Q}_p(\mu_{m'})|\mathbb{Q}_p$, but also in the norm group of $\mathbb{Q}_p(\mu_{p^{n_p}})|\mathbb{Q}_p$, according to chap. V, (1.8). This means, by § 3, that every idèle in $I_{\mathbb{Q}}^{\mathfrak{m}}$ is a norm of some idèle of $\mathbb{Q}(\mu_m)$. Thus $C_{\mathbb{Q}}^{\mathfrak{m}} \subseteq N C_{\mathbb{Q}(\mu_m)}$. On the other hand, $C_{\mathbb{Q}}/C_{\mathbb{Q}}^{\mathfrak{m}} \cong (\mathbb{Z}/m\mathbb{Z})^*$ by (1.10), and therefore

$$\left(C_{\mathbb{Q}} : C_{\mathbb{Q}}^{\mathfrak{m}}\right) = \left[\mathbb{Q}(\mu_m) : \mathbb{Q}\right] = \left(C_{\mathbb{Q}} : N C_{\mathbb{Q}(\mu_m)}\right),$$

so that $C_{\mathbb{Q}}^{\mathfrak{m}} = N C_{\mathbb{Q}(\mu_m)}$, and this proves the claim. □

According to this proposition, one may view the general ray class fields $K^{\mathfrak{m}}|K$ as analogues of the cyclotomic fields $\mathbb{Q}(\mu_m)|\mathbb{Q}$. Nonetheless, they are not made to take over the important rôle of the latter because all we know about them is that they exist, but not how to generate them. In the case of local fields things were different. There the analogues of the ray class fields were the Lubin-Tate extensions which could be generated by the division points of formal groups – a fact that carries a long way (see chap. V, § 5). This local discovery does, however, originate from the problem of generating global class fields, which will be discussed at the end of this section.

Note in passing that the above proposition gives another proof of the theorem of Kronecker and Weber (see chap. V, (1.10)) to the effect that every finite abelian extension $L|\mathbb{Q}$ is contained in a field $\mathbb{Q}(\mu_m)|\mathbb{Q}$, because by (1.8) the norm group $N_{L|\mathbb{Q}}C_L$ lies in some congruence subgroup $C_{\mathbb{Q}}^{\mathfrak{m}}$, $\mathfrak{m} = (m)$, so that $L \subseteq \mathbb{Q}(\mu_m)$.

Among all abelian extensions of K, the ray class field mod 1 occupies a special place. It is called the **big Hilbert class field** and has Galois group

$$G(K^1|K) \cong Cl_K^1 .$$

By (1.11), the group Cl_K^1 is linked to the ordinary ideal class group by the exact sequence

$$1 \longrightarrow o^*/o_+^* \longrightarrow \prod_{\mathfrak{p}\text{ real}} \mathbb{R}^*/\mathbb{R}_+^* \longrightarrow Cl_K^1 \longrightarrow Cl_K \longrightarrow 1.$$

The big Hilbert class field has conductor $\mathfrak{f} = 1$ and may therefore be characterized by (6.6) in the following way.

(6.8) Proposition. *The big Hilbert class field is the maximal unramified abelian extension of K.*

Since the infinite places are always unramified, this means that all prime ideals are unramified. The **Hilbert class field**, or more precisely, the "small Hilbert class field", is defined to be the maximal unramified abelian extension $H|K$ in which all infinite places split completely, i.e., the real places stay real. It satisfies the

(6.9) Proposition. *The Galois group of the small Hilbert class field $H|K$ is canonically isomorphic to the ideal class group:*

$$G(H|K) \cong Cl_K .$$

In particular, the degree $[H : K]$ is the class number h_K of K.

Proof: We consider the big Hilbert class field $K^1|K$ and, for every infinite place $\mathfrak{p}$, the commutative diagram (see (5.6))

$$\begin{array}{ccc} K_{\mathfrak{p}}^* & \xrightarrow{(\ ,K_{\mathfrak{p}}^1|K_{\mathfrak{p}})} & G(K_{\mathfrak{p}}^1|K_{\mathfrak{p}}) \\ {\scriptstyle (\ )}\downarrow & & \downarrow \\ I_K/I_K^1K^* & \xrightarrow{(\ ,K^1|K)} & G(K^1|K). \end{array}$$

The small Hilbert class field $H|K$ is the fixed field of the subgroup G_∞ generated by all $G(K^1_\mathfrak{p}|K_\mathfrak{p})$, $\mathfrak{p}|\infty$. Under $(\ ,K^1|K)$ this is the image of

$$\Big(\prod_{\mathfrak{p}|\infty} K_\mathfrak{p}^*\Big)\, I_K^1 K^*/I_K^1 K^* = I_K^{S_\infty} K^*/I_K^1 K^*,$$

where $I_K^{S_\infty} = \prod_{\mathfrak{p}|\infty} K_\mathfrak{p}^* \times \prod_{\mathfrak{p}\nmid\infty} U_\mathfrak{p}$. Therefore by (1.3),

$$G(H|K) = G(K^1|K)/G_\infty \cong I_K/I_K^{S_\infty}K^* \cong Cl_K\,. \qquad \square$$

Remark: The small Hilbert class field is in general not a ray class field in terms of the theory developed here. But it is in many other textbooks where ray class groups and ray class fields are defined differently (see for instance [107]). This other theory is obtained by equipping all number fields with the Minkowski metric

$$\langle x,y\rangle_K = \sum_\tau \alpha_\tau x_\tau \bar{y}_\tau \qquad \big(\tau \in \operatorname{Hom}(K,\mathbb{C})\big)\,,$$

$\alpha_\tau = 1$ if $\tau = \bar\tau$, $\alpha_\tau = \frac{1}{2}$ if $\tau \neq \bar\tau$. A ray class group can then be attached to any *replete* module

$$\mathfrak{m} = \prod_\mathfrak{p} \mathfrak{p}^{n_\mathfrak{p}},$$

where $n_\mathfrak{p} \in \mathbb{Z}$, $n_\mathfrak{p} \geq 0$, and $n_\mathfrak{p} = 0$ or $= 1$ if $\mathfrak{p}|\infty$. The groups $U_\mathfrak{p}^{(n_\mathfrak{p})}$ attached to the metrized number field $(K, \langle\ ,\ \rangle_K)$ are defined by

$$U_\mathfrak{p}^{(n_\mathfrak{p})} = \begin{cases} 1+\mathfrak{p}^{n_\mathfrak{p}}, & \text{for } n_\mathfrak{p} > 0, \text{ and } U_\mathfrak{p} \text{ for } n_\mathfrak{p} = 0, \text{ if } \mathfrak{p} \nmid \infty, \\ \mathbb{R}^*, & \text{if } \mathfrak{p} \text{ is real and } n_\mathfrak{p} = 0, \\ \mathbb{R}_+^*, & \text{if } \mathfrak{p} \text{ is real and } n_\mathfrak{p} = 1, \\ \mathbb{C}^* = K_\mathfrak{p}^*, & \text{if } \mathfrak{p} \text{ is complex.} \end{cases}$$

The *congruence subgroup* mod $\mathfrak{m}$ of $(K, \langle\ ,\ \rangle_K)$ is then the subgroup $C_K^\mathfrak{m} = I_K^\mathfrak{m} K^*/K^*$ of C_K formed with the group

$$I_K^\mathfrak{m} = \prod_\mathfrak{p} U_\mathfrak{p}^{(n_\mathfrak{p})},$$

and the factor group $C_K/C_K^\mathfrak{m}$ is the *ray class group* mod $\mathfrak{m}$. The *ray class field* mod $\mathfrak{m}$ of $(K, \langle\ ,\ \rangle_K)$ is again the class field of K corresponding to the group $C_K^\mathfrak{m} \subseteq C_K$. As explained in chap. III, § 3, the infinite places $\mathfrak{p}$ have to be considered as *ramified* in an extension $L|K$ if $L_\mathfrak{p} \neq K_\mathfrak{p}$. Likewise, the *conductor* of an abelian extension $L|K$, i.e., the gcd of all modules $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$ such that $C_K^{\mathfrak{m}} \subseteq N_{L|K} C_L$, is the replete ideal

$$\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}},$$

where now for an infinite place $\mathfrak{p}$, we have $\mathfrak{f}_{\mathfrak{p}} = \mathfrak{p}^{n_{\mathfrak{p}}}$ with $n_{\mathfrak{p}} = 0$ if $L_{\mathfrak{p}} = K_{\mathfrak{p}}$, and $n_{\mathfrak{p}} = 1$ if $L_{\mathfrak{p}} \neq K_{\mathfrak{p}}$. Corollary (6.6) then continues to hold: a place $\mathfrak{p}$ is ramified in L if and only if $\mathfrak{p}$ occurs in the conductor $\mathfrak{f}$.

This entails the following modifications of the above theory, as far as ray class fields are concerned. The ray class field mod 1 is the *small* Hilbert class field. It is now the maximal abelian extension of K which is unramified at *all* places. The big Hilbert class field is the ray class field for the module $\mathfrak{m} = \prod_{\mathfrak{p}|\infty} \mathfrak{p}$. In the case of the base field $\mathbb{Q}$, the field $\mathbb{Q}(\zeta)$ of m-th roots of unity is the ray class field mod mp_∞, where p_∞ is the infinite place. The ray class field for the module m becomes the maximal real subextension $\mathbb{Q}(\zeta + \zeta^{-1})$, which was not a ray class field before. This is the theory one finds in the textbooks alluded to above. It corresponds to the number fields with the Minkowski metric. The theory of ray class fields according to the treatment of this book is forced upon us already by the choice of the standard metric $\langle x, y\rangle = \sum_\tau x_\tau \bar{y}_\tau$ on $K_{\mathbb{R}}$ taken in chap. I, §5. It is compatible with the Riemann-Roch theory of chap. III, and has the advantage of being simpler.

Over the field $\mathbb{Q}$, the ray class field mod (m) can be generated, according to (6.7), by the m-th roots of unity, i.e., by special values of the exponential function $e^{2\pi i z}$. The question suggested by this observation is whether one may construct the abelian extensions of an arbitrary number field in a similarly concrete way, via special values of analytic functions. This was the historic origin of the notion of class field. A completely satisfactory answer to this question has been given only in the case of an imaginary quadratic field K. The results for this case are subsumed under the name of **Kronecker's Jugendtraum** (Kronecker's dream of his youth). We will briefly describe them here. For the proofs, which presuppose an in-depth knowledge of the theory of *elliptic curves*, we have to refer to [96] and [28].

An elliptic curve is given as the quotient $E = \mathbb{C}/\Gamma$ of $\mathbb{C}$ by a complete lattice $\Gamma = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ in $\mathbb{C}$. This is a torus which receives the structure of an algebraic curve via the **Weierstrass $\wp$-function**

$$\wp(z) = \wp_\Gamma(z) = \frac{1}{z^2} + \sum_{\omega\in\Gamma'} \Big[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\Big],$$

where $\Gamma' = \Gamma \smallsetminus \{0\}$. $\wp(z)$ is a meromorphic doubly periodic function, i.e.,

$$\wp(z+\omega) = \wp(z) \quad \text{for all} \quad \omega \in \Gamma,$$

and it satisfies, along with its derivative $\wp'(z)$, an identity

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 .$$

The constants g_2, g_3 only depend on the lattice Γ, and are given by $g_2 = g_2(\Gamma) = 60\sum_{\omega\neq 0}\frac{1}{\omega^4}$, $g_3 = g_3(\Gamma) = 140\sum_{\omega\neq 0}\frac{1}{\omega^6}$. $\wp$ and $\wp'$ may thus be interpreted as functions on $\mathbb{C}/\Gamma$. If one takes away the finite set $S \subseteq \mathbb{C}/\Gamma$ of poles, one gets a bijection

$$\mathbb{C}/\Gamma \smallsetminus S \xrightarrow{\sim} \left\{(x,y) \in \mathbb{C}^2 \mid y^2 = 4x^3 - g_2x - g_3\right\}, \quad z \longmapsto \left(\wp(z), \wp'(z)\right),$$

onto the affine algebraic curve in $\mathbb{C}^2$ given by the equation $y^2 = 4x^3 - g_2x - g_3$. This gives the torus $\mathbb{C}/\Gamma$ the structure of an algebraic curve E over $\mathbb{C}$ of genus 1. An important rôle is played by the **j-invariant**

$$j(E) = j(\Gamma) = \frac{2^6 3^3 g_2^3}{\Delta} \qquad \text{with} \quad \Delta = g_2^3 - 27g_3^2 .$$

It determines the elliptic curve E up to isomorphism. Writing generators ω_1, ω_2 of Γ in such an order that $\tau = \omega_1/\omega_2$ lies in the upper half-plane $\mathbb{H}$, then $j(E)$ becomes the value $j(\tau)$ of a **modular function**, i.e., of a holomorphic function j on $\mathbb{H}$ which is invariant under the substitution $\tau \mapsto \dfrac{a\tau + b}{c\tau + d}$ for every matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$.

Now let $K \subseteq \mathbb{C}$ be an imaginary quadratic number field. Then the ring $\mathcal{O}_K$ of integers forms a lattice in $\mathbb{C}$, and more generally, any ideal $\mathfrak{a}$ of $\mathcal{O}_K$ does as well. The tori $\mathbb{C}/\mathfrak{a}$ constructed in this way are elliptic curves with *complex multiplication.* This means the following. An endomorphism of an elliptic curve $E = \mathbb{C}/\Gamma$ is given as multiplication by a complex number z such that $z\Gamma \subseteq \Gamma$. Generically, one has $\operatorname{End}(E) = \mathbb{Z}$. If this is not the case, then $\operatorname{End}(E) \otimes \mathbb{Q}$ is necessarily an imaginary quadratic number field K, and one says that this is an elliptic curve with complex multiplication. The curves $\mathbb{C}/\mathfrak{a}$ are obviously of this kind.

The consequences of these analytic investigations for class field theory are the following.

(6.10) Theorem. *Let K be an imaginary quadratic number field and $\mathfrak{a}$ an ideal of $\mathcal{O}_K$. Then one has:*

(i) *The j-invariant $j(\mathfrak{a})$ of $\mathbb{C}/\mathfrak{a}$ is an algebraic integer which depends only on the ideal class $\mathfrak{K}$ of $\mathfrak{a}$, and will therefore be denoted by $j(\mathfrak{K})$.*

(ii) *Every $j(\mathfrak{a})$ generates the Hilbert class field over K.*

(iii) *If* $\mathfrak{a}_1, \ldots, \mathfrak{a}_h$ *are representatives of the ideal class group* Cl_K, *then the numbers* $j(\mathfrak{a}_i)$ *are conjugate to one another over* K.

(iv) *For almost all prime ideals* $\mathfrak{p}$ *of* K *one has*

$$\varphi_{\mathfrak{p}} j(\mathfrak{a}) = j(\mathfrak{p}^{-1}\mathfrak{a}),$$

where $\varphi_{\mathfrak{p}} \in G(K(j(\mathfrak{a}))|K)$ *is the Frobenius automorphism of a prime ideal* $\mathfrak{P}$ *of* $K(j(\mathfrak{a}))$ *above* $\mathfrak{p}$.

Note that for a totally imaginary field K there is no difference between big and small Hilbert class field. In order to go beyond the Hilbert class field, i.e., the ray class field mod 1, to the ray class fields for arbitrary modules $\mathfrak{m} \neq 1$, we form, for any lattice $\Gamma \subseteq \mathbb{C}$, the **Weber function**

$$\tau_\Gamma(z) = \begin{cases} -2^7 3^5 \frac{g_2 g_3}{\Delta} \wp_\Gamma(z), & \text{if } g_2 g_3 \neq 0, \\ -2^9 3^6 \frac{g_3}{\Delta} \wp_\Gamma^3(z), & \text{if } g_2 = 0, \\ 2^8 3^4 \frac{g_2^2}{\Delta} \wp_\Gamma^2(z), & \text{if } g_3 = 0. \end{cases}$$

Let $\mathfrak{K} \in Cl_K$ be an ideal class chosen once and for all. We denote by $\mathfrak{K}^*$ the classes in the ray class group $Cl_K^{\mathfrak{m}} = J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ which under the homomorphism

$$Cl_K^{\mathfrak{m}} \longrightarrow Cl_K$$

are sent to the ideal class $(\mathfrak{m})\mathfrak{K}^{-1}$. Let $\mathfrak{a}$ be an ideal in $\mathfrak{K}$, and let $\mathfrak{b}$ be an integral ideal in $\mathfrak{K}^*$. Then $\mathfrak{a}\mathfrak{b}\mathfrak{m}^{-1} = (a)$ is a principal ideal. The value $\tau_{\mathfrak{a}}(a)$ only depends on the class $\mathfrak{K}^*$, not on the choice of $\mathfrak{a}, \mathfrak{b}$ and a. It will be denoted by

$$\tau(\mathfrak{K}^*) = \tau_{\mathfrak{a}}(a).$$

With these conventions we then have the

(6.11) Theorem. (i) *The invariants* $\tau(\mathfrak{K}_1^*), \tau(\mathfrak{K}_2^*), \ldots,$ *for a fixed ideal class* $\mathfrak{K}$, *are distinct algebraic numbers which are conjugate over the Hilbert class field* $K^1 = K(j(\mathfrak{K}))$.

(ii) *For an arbitrary* $\mathfrak{K}^*$, *the field* $K(j(\mathfrak{K}), \tau(\mathfrak{K}^*))$ *is the ray class field* mod $\mathfrak{m}$ *over* K:

$$K^{\mathfrak{m}} = K\big(j(\mathfrak{K}), \tau(\mathfrak{K}^*)\big).$$

Exercise 1. Let $K^1|K$ be the big, and $H|K$ the small Hilbert class field. Then $G(K^1|H) \cong (\mathbb{Z}/2\mathbb{Z})^{r-t}$, where r is the number of real places, and $2^t = (\mathcal{O}^* : \mathcal{O}^*_+)$.

Exercise 2. Let $d > 0$ be squarefree, and $K = \mathbb{Q}(\sqrt{d})$. Let ε be a totally positive fundamental unit of K. Then one has $[K^1 : H] = 1$ or $= 2$, according as $N_{K|\mathbb{Q}}(\varepsilon) = -1$ or $= 1$.

Exercise 3. The group $(C_K)^n = (I_K)^n K^*/K^*$ is the intersection of the norm groups $N_{L|K}C_L$ of all abelian extensions $L|K$ of exponent n.

Exercise 4. (i) For a number field K, local Tate duality (see chap. V, § 1, exercise 2) yields a non-degenerate pairing

$$(*) \qquad \prod_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mathbb{Z}/n\mathbb{Z}) \times \prod_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mu_n) \to \mathbb{Z}/n\mathbb{Z}$$

of locally compact groups, where the restricted products are taken with respect to the subgroups $H^1_{nr}(K_{\mathfrak{p}}, \mathbb{Z}/n\mathbb{Z})$, resp. $H^1_{nr}(K_{\mathfrak{p}}, \mu_n)$. For $\chi = (\chi_{\mathfrak{p}})$ in the first and $\alpha = (\alpha_{\mathfrak{p}})$ in the second product, it is given by

$$(\chi, \alpha) = \sum_{\mathfrak{p}} \chi_{\mathfrak{p}}(\alpha_{\mathfrak{p}}, \bar{K}_{\mathfrak{p}}|K_{\mathfrak{p}}).$$

(ii) If $L|K$ is a finite extension, then one has a commutative diagram

$$\begin{array}{ccccc}
\prod\limits_{\mathfrak{P}} H^1(L_{\mathfrak{P}}, \mathbb{Z}/n\mathbb{Z}) & \times & \prod\limits_{\mathfrak{P}} H^1(L_{\mathfrak{P}}, \mu_n) & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \\
\uparrow & & \big\downarrow{\scriptstyle N_{L|K}} & & \| \\
\prod\limits_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mathbb{Z}/n\mathbb{Z}) & \times & \prod\limits_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mu_n) & \longrightarrow & \mathbb{Z}/n\mathbb{Z}.
\end{array}$$

(iii) The images of

$$H^1(K, \mathbb{Z}/n\mathbb{Z}) \to \prod_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mathbb{Z}/n\mathbb{Z})$$

and

$$H^1(K, \mu_n) \to \prod_{\mathfrak{p}} H^1(K_{\mathfrak{p}}, \mu_n)$$

are mutual orthogonal complements with respect to the pairing $(*)$.

Hint for (iii): The cokernel of the second map is $C_K/(C_K)^n$, and one has $H^1(K, \mathbb{Z}/n\mathbb{Z}) = \operatorname{Hom}(G(L|K), \mathbb{Z}/n\mathbb{Z})$, where $L|K$ is the maximal abelian extension of exponent n.

Exercise 5 (Global Tate Duality). Show that the statements of exercise 4 extend to an arbitrary finite G_K-module A instead of $\mathbb{Z}/n\mathbb{Z}$, and $A' = \operatorname{Hom}(A, \bar{K}^*)$ instead of μ_n.

Hint: Use exercises 4–8 of chap. IV, § 3, and exercise 4 of chap. V, § 1.

Exercise 6. If S is a finite set of places of K, then the map

$$H^1(K, \mathbb{Z}/n\mathbb{Z}) \to \prod_{\mathfrak{p} \in S} H^1(K_{\mathfrak{p}}, \mathbb{Z}/n\mathbb{Z})$$

is surjective if and only if the map

$$H^1(K, \mu_n) \to \prod_{\mathfrak{p} \notin S} H^1(K_{\mathfrak{p}}, \mu_n)$$

is injective. This is the case in particular if either the extension $K(\mu_{2^\nu})|K$ is cyclic, $n = 2^\nu m$, $(m, 2) = 1$, or if S does not contain all places $\mathfrak{p}|2$ which are nonsplit in $K(\mu_{2^\nu})$ (see § 1, exercise 2).

Exercise 7 (Theorem of *Grunwald***).** If the last condition of exercise 6 is satisfied for the triple (K, n, S), then, given cyclic extensions $L_\mathfrak{p}|K_\mathfrak{p}$ for $\mathfrak{p} \in S$, there always exists a cyclic extension $L|K$ which has $L_\mathfrak{p}|K_\mathfrak{p}$ as a completion for $\mathfrak{p} \in S$, and which satisfies the identity of degrees

$$[L : K] = \mathrm{scm}\{[L_\mathfrak{p} : K_\mathfrak{p}]\}$$

(see also [10], chap. X, § 2).

Note: Let G be a finite group of order prime to $\#\mu(K)$, let S be a finite set of places, and let $L_\mathfrak{p}|K_\mathfrak{p}$, $\mathfrak{p} \in S$, be given Galois extensions whose Galois groups $G_\mathfrak{p}$ can be embedded into G. Then there exists a Galois extension $L|K$ which on the one hand has Galois group isomorphic to G, and which on the other hand has the given extensions $L_\mathfrak{p}|K_\mathfrak{p}$ as completions (see [109]).

§ 7. The Ideal-Theoretic Version of Class Field Theory

Class field theory has found its idèle-theoretic formulation only after it had been completed in the language of ideals. From the very start, it was guided by the desire to classify all abelian extensions of a number field K. But at first, instead of the idèle class group C_K, there was only the ideal class group Cl_K at hand to do this, along with its subgroups. In terms of the insights that we have gained in the preceding section, this means the restriction to the subfields of the Hilbert class field, i.e., to the *unramified* abelian extensions of K. If the base field is $\mathbb{Q}$, this restriction is of course radical, for $\mathbb{Q}$ has no unramified extensions at all by Minkowski's theorem. But over $\mathbb{Q}$, we naturally encounter the cyclotomic fields $\mathbb{Q}(\mu_m)|\mathbb{Q}$ with their familiar isomorphisms $G(\mathbb{Q}(\mu_m)|\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*$. *Heinrich Weber* realized, as was already mentioned, that the groups Cl_K and $(\mathbb{Z}/m\mathbb{Z})^*$ are – with a grain of salt – only different instances of a common concept, that of a ray class group, which he defined in an ideal-theoretic way as the quotient group

$$Cl_K^\mathfrak{m} = J_K^\mathfrak{m}/P_K^\mathfrak{m}$$

of all ideals relatively prime to a given module $\mathfrak{m}$, by the principal ideals (α) with $\alpha \equiv 1 \bmod \mathfrak{m}$, and α totally positive. He conjectured that this group $Cl_K^\mathfrak{m}$, along with its subgroups, would do the same for the subextensions of a "ray class field" $K^\mathfrak{m}|K$ (which at first was only postulated to exist) as the ideal class group Cl_K and its subgroups did for the subfields of the Hilbert class field. Moreover, he stated the hypothesis that every abelian extension ought to be captured by such a ray class field, as was suggested by the case where the base field is $\mathbb{Q}$, whose abelian extensions are all contained in cyclotomic fields $\mathbb{Q}(\mu_m)|\mathbb{Q}$ by the Kronecker-Weber theorem. After the seminal work of the Austrian mathematician *Philipp Furtwängler* [44], these conjectures were confirmed by the Japanese arithmetician *Teiji Takagi* (1875–1960), and cast by *Emil Artin* (1898–1962) into a definite, canonical form.

The idèle-theoretic language introduced by *Chevalley* brought the simplification that the idèle class group C_K encapsulated all abelian extensions of $L|K$ at once, avoiding choosing a module $\mathfrak{m}$ every time such an extension was given, in order to accommodate it into the ray class field $K^{\mathfrak{m}}|K$, and thereby make it amenable to class field theory. The classical point of view can be vindicated in terms of the idèle-theoretic version by looking at congruence subgroups $C_K^{\mathfrak{m}}$ in C_K, which define the ray class fields $K^{\mathfrak{m}}|K$. Their subfields correspond, according to the new point of view, to the groups between $C_K^{\mathfrak{m}}$ and C_K, and hence, in view of the isomorphism

$$C_K/C_K^{\mathfrak{m}} \cong Cl_K^{\mathfrak{m}},$$

to the subgroups of the ray class group $Cl_K^{\mathfrak{m}}$.

In what follows, we want to deduce the classical, ideal-theoretic version of global class field theory from the idèle-theoretic one. This is not only an obligation towards history, but a factual necessity that is forced upon us by the numerous applications of the more elementary and more immediately accessible ideal groups.

Let $L|K$ be an abelian extension, and let $\mathfrak{p}$ be an unramified prime ideal of K and $\mathfrak{P}$ a prime ideal of L lying above $\mathfrak{p}$. The decomposition group $G(L_{\mathfrak{P}}|K_{\mathfrak{p}}) \subseteq G(L|K)$ is then generated by the classical **Frobenius automorphism**

$$\varphi_{\mathfrak{p}} = (\pi_{\mathfrak{p}}, L_{\mathfrak{P}}|K_{\mathfrak{p}}),$$

where $\pi_{\mathfrak{p}}$ is a prime element of $K_{\mathfrak{p}}$. As an automorphism of L, $\varphi_{\mathfrak{p}}$ is obviously characterized by the congruence

$$\varphi_{\mathfrak{p}} a \equiv a^q \bmod \mathfrak{P} \quad \text{for all} \quad a \in \mathcal{O}_L$$

where q is the number of elements in the residue class field of $\mathfrak{p}$. We put

$$\varphi_{\mathfrak{p}} =: \left(\frac{L|K}{\mathfrak{p}}\right).$$

Now let $\mathfrak{m}$ be a module of K such that L lies in the ray class field mod $\mathfrak{m}$. Such a module is called an **module of definition** for L. Since by (6.6) each prime ideal $\mathfrak{p} \nmid \mathfrak{m}$ is unramified in L, we get a canonical homomorphism

$$\left(\frac{L|K}{\quad}\right) : J_K^{\mathfrak{m}} \longrightarrow G(L|K)$$

from the group $J_K^{\mathfrak{m}}$ of all ideals of K which are relatively prime to $\mathfrak{m}$ by putting, for any ideal $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$:

$$\left(\frac{L|K}{\mathfrak{a}}\right) = \prod_{\mathfrak{p}} \left(\frac{L|K}{\mathfrak{p}}\right)^{\nu_{\mathfrak{p}}}.$$

$\left(\frac{L|K}{\mathfrak{a}}\right)$ is called the **Artin symbol.** If $\mathfrak{p} \in J_K^{\mathfrak{m}}$ is a prime ideal and $\pi_{\mathfrak{p}}$ a prime element of $K_{\mathfrak{p}}$, then clearly

$$\left(\frac{L|K}{\mathfrak{p}}\right) = \left(\langle \pi_{\mathfrak{p}} \rangle, L|K\right),$$

if $\langle \pi_{\mathfrak{p}} \rangle \in C_K$ denotes the class of the idèle $(\dots, 1, 1, \pi_{\mathfrak{p}}, 1, 1, \dots)$.

The relation between the idèle-theoretic and the ideal-theoretic formulation of the Artin reciprocity law is now provided by the following theorem.

(7.1) Theorem. *Let $L|K$ be an abelian extension, and let $\mathfrak{m}$ be a module of definition for it. Then the Artin symbol induces a surjective homomorphism*

$$\left(\frac{L|K}{\quad}\right) : Cl_K^{\mathfrak{m}} \longrightarrow G(L|K)$$

with kernel $H^{\mathfrak{m}}/P_K^{\mathfrak{m}}$, where $H^{\mathfrak{m}} = (N_{L|K} J_L^{\mathfrak{m}})P_K^{\mathfrak{m}}$, and we have an exact commutative diagram

$$\begin{tikzcd}
1 \arrow[r] & N_{L|K}C_L \arrow[r] \arrow[d] & C_K \arrow[r, "(\ {,}L|K)"] \arrow[d] & G(L|K) \arrow[r] \arrow[d, "\mathrm{id}"] & 1 \\
1 \arrow[r] & H^{\mathfrak{m}}/P_K^{\mathfrak{m}} \arrow[r] & Cl_K^{\mathfrak{m}} \arrow[r, "\left(\frac{L|K}{\ }\right)"] & G(L|K) \arrow[r] & 1.
\end{tikzcd}$$

Proof: In §1, we obtained the isomorphism $(\ ) : C_K/C_K^{\mathfrak{m}} \to Cl_K^{\mathfrak{m}} = J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ by sending an idèle $\alpha = (\alpha_{\mathfrak{p}})$ to the ideal $(\alpha) = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}(\alpha_{\mathfrak{p}})}$. This isomorphism yields a commutative diagram

$$\begin{tikzcd}
C_K/C_K^{\mathfrak{m}} \arrow[r, "(\ {,}L|K)"] \arrow[d, "(\ )"'] & G(L|K) \arrow[d, "\mathrm{id}"] \\
Cl_K^{\mathfrak{m}} \arrow[r, "f"] & G(L|K),
\end{tikzcd}$$

and we show that f is given by the Artin symbol.

Let $\mathfrak{p}$ be a prime ideal not dividing $\mathfrak{m}$, $\pi_\mathfrak{p}$ a prime element of $K_\mathfrak{p}$, and $c \in C_K/C_K^\mathfrak{m}$ the class of the idèle $\langle\pi_\mathfrak{p}\rangle = (\dots,1,1,\pi_\mathfrak{p},1,1,\dots)$. Then $(c) = \mathfrak{p} \bmod P_K^\mathfrak{m}$ and

$$f\big((c)\big) = (c, L|K) = \big(\langle\pi_\mathfrak{p}\rangle, L|K\big) = \Big(\frac{L|K}{\mathfrak{p}}\Big).$$

This shows that $f : J_K^\mathfrak{m}/P_K^\mathfrak{m} \to G(L|K)$ is induced by the Artin symbol $\big(\frac{L|K}{\ }\big) : J_K^\mathfrak{m} \to G(L|K)$, and that it is surjective.

It remains to show that the image of $N_{L|K}C_L$ under the map $(\ ) : C_K \to J_K^\mathfrak{m}/P_K^\mathfrak{m}$ is the group $H^\mathfrak{m}/P_K^\mathfrak{m}$. We view the module $\mathfrak{m} = \prod_{\mathfrak{p}\nmid\infty}\mathfrak{p}^{n_\mathfrak{p}}$ as a module of L by substituting for each prime ideal $\mathfrak{p}$ of K the product $\mathfrak{p} = \prod_{\mathfrak{P}|\mathfrak{p}}\mathfrak{P}^{e_{\mathfrak{P}|\mathfrak{p}}}$. As in the proof of (1.9), we then get $C_L = I_L^{(\mathfrak{m})}L^*/L^*$, where $I_L^{(\mathfrak{m})} = \{\alpha \in I_L \mid \alpha_\mathfrak{P} \in U_\mathfrak{P}^{(e_{\mathfrak{P}|\mathfrak{p}}n_\mathfrak{p})} \text{ for } \mathfrak{P}|\mathfrak{m}\infty\}$. The elements of

$$N_{L|K}C_L = N_{L|K}(I_L^{(\mathfrak{m})})K^*/K^*$$

are the classes of norm idèles $N_{L|K}(\alpha)$, for $\alpha \in I_L^{(\mathfrak{m})}$. As

$$N_{L|K}(\alpha)_\mathfrak{p} = \prod_{\mathfrak{P}|\mathfrak{p}} N_{L_\mathfrak{P}|K_\mathfrak{p}}(\alpha_\mathfrak{P})$$

(see (2.2)), and since $v_\mathfrak{p}(N_{L_\mathfrak{P}|K_\mathfrak{p}}(\alpha_\mathfrak{P})) = f_{\mathfrak{P}|\mathfrak{p}}v_\mathfrak{P}(\alpha_\mathfrak{P})$ (see chap. III, (1.2)), the idèle $N_{L|K}(\alpha)$ is mapped by $(\ )$ to the ideal

$$(N_{L|K}(\alpha)) = \prod_{\mathfrak{p}\nmid\infty}\prod_{\mathfrak{P}|\mathfrak{p}}\mathfrak{p}^{f_{\mathfrak{P}|\mathfrak{p}}v_\mathfrak{P}(\alpha_\mathfrak{P})} = N_{L|K}\Big(\prod_{\mathfrak{P}\nmid\infty}\mathfrak{P}^{v_\mathfrak{P}(\alpha_\mathfrak{P})}\Big).$$

Therefore the image of $N_{L|K}C_L$ under the homomorphism $(\ ) : C_K \to J_K^\mathfrak{m}/P_K^\mathfrak{m}$ is precisely the group $(N_{L|K}J_L^\mathfrak{m})P_K^\mathfrak{m}/P_K^\mathfrak{m}$, q.e.d. □

(7.2) Corollary. *The Artin symbol* $\big(\frac{L|K}{\mathfrak{a}}\big)$, *for* $\mathfrak{a} \in J_K^\mathfrak{m}$, *only depends on the class* $\mathfrak{a} \bmod P_K^\mathfrak{m}$. *It defines an isomorphism*

$$\Big(\frac{L|K}{\ }\Big) : J_K^\mathfrak{m}/H^\mathfrak{m} \xrightarrow{\ \sim\ } G(L|K).$$

The group $H^\mathfrak{m} = (N_{L|K}J_L^\mathfrak{m})P_K^\mathfrak{m}$ is called the "*ideal group defined* mod $\mathfrak{m}$" belonging to the extension $L|K$. From the existence theorem (6.1), we see that the correspondence $L \mapsto H^\mathfrak{m}$ is $1-1$ between subextensions of the ray class field mod $\mathfrak{m}$ and subgroups of $J_K^\mathfrak{m}$ containing $P_K^\mathfrak{m}$.

The most important consequence of theorem (7.1) is a precise analysis of the kind of decomposition of any unramified prime ideal $\mathfrak{p}$ in an abelian extension $L|K$. It can be immediately read off the ideal group $H^\mathfrak{m} \subseteq J_K^\mathfrak{m}$ which determines the field L as class field.

(7.3) Theorem (Decomposition Law). *Let $L|K$ be an abelian extension of degree n, and let $\mathfrak{p}$ be an unramified prime ideal. Let $\mathfrak{m}$ be a module of definition for $L|K$ that is not divisible by $\mathfrak{p}$ (for instance the conductor), and let $H^{\mathfrak{m}}$ be the corresponding ideal group.*

If f is the order of $\mathfrak{p}$ mod *$H^{\mathfrak{m}}$ in the class group $J_K^{\mathfrak{m}}/H^{\mathfrak{m}}$, i.e., the smallest positive integer such that*

$$\mathfrak{p}^f \in H^{\mathfrak{m}},$$

then $\mathfrak{p}$ decomposes in L into a product

$$\mathfrak{p} = \mathfrak{P}_1 \cdots \mathfrak{P}_r$$

of $r = n/f$ distinct prime ideals of degree f over $\mathfrak{p}$.

Proof: Let $\mathfrak{p} = \mathfrak{P}_1 \cdots \mathfrak{P}_r$ be the prime decomposition of $\mathfrak{p}$ in L. Since $\mathfrak{p}$ is unramified, the $\mathfrak{P}_i$ are all distinct and have the same degree f. This degree is the order of the decomposition group of $\mathfrak{P}_i$ over K, i.e., the order of the Frobenius automorphism $\varphi_{\mathfrak{p}} = \left(\frac{L|K}{\mathfrak{p}}\right)$. In view of the isomorphism $J_K^{\mathfrak{m}}/H^{\mathfrak{m}} \cong G(L|K)$, this is also the order of $\mathfrak{p}$ mod $H^{\mathfrak{m}}$ in $J_K^{\mathfrak{m}}/H^{\mathfrak{m}}$. This finishes the proof. □

The theorem shows in particular that the prime ideals which split completely are precisely those contained in the ideal group $H^{\mathfrak{f}}$, if $\mathfrak{f}$ is the conductor of $L|K$.

Let us highlight two special cases. If the base field is $K = \mathbb{Q}$ and we look at the cyclotomic field $\mathbb{Q}(\mu_m)|\mathbb{Q}$, the conductor is the module $\mathfrak{m} = (m)$, and the ideal group corresponding to $\mathbb{Q}(\mu_m)$ in $J_{\mathbb{Q}}^{\mathfrak{m}}$ is the group $P_{\mathbb{Q}}^{\mathfrak{m}}$. As $J_{\mathbb{Q}}^{\mathfrak{m}}/P_{\mathbb{Q}}^{\mathfrak{m}} \cong (\mathbb{Z}/m\mathbb{Z})^*$ (see (1.10)), we obtain for the decomposition of rational primes $p \nmid m$, the law which we had already deduced in chap. I, (10.4), and in particular the fact that the prime numbers which split completely are characterized by

$$p \equiv 1 \bmod m.$$

In the case of the Hilbert class field $L|K$, i.e., of the field inside the ray class field mod 1 in which the infinite places split completely, the corresponding ideal group $H \subseteq J_K^1 = J_K$ is the group P_K of principal ideals (see (6.9)). This gives us the strikingly simple

(7.4) Corollary. *The prime ideals of K which split completely in the Hilbert class field are precisely the principal prime ideals.*

Another highly remarkable property of the Hilbert class field is expressed by the following theorem, known as the **principal ideal theorem**.

(7.5) Theorem. *In the Hilbert class field every ideal $\mathfrak{a}$ of K becomes a principal ideal.*

Proof: Let $K_1|K$ be the Hilbert class field of K and let $K_2|K_1$ be the Hilbert class field of K_1. We have to show that the canonical homomorphism

$$J_K/P_K \longrightarrow J_{K_1}/P_{K_1}$$

is trivial. By chap. IV, (5.9), we have a commutative diagram

$$\begin{array}{ccccc} J_{K_1}/P_{K_1} & \cong & C_{K_1}/N_{K_2|K_1}C_{K_2} & \cong & G(K_2|K_1) \\ \uparrow & & \uparrow{\scriptstyle i} & & \uparrow{\scriptstyle \mathrm{Ver}} \\ J_K/P_K & \cong & C_K/N_{K_1|K}C_{K_1} & \cong & G(K_1|K), \end{array}$$

where i is induced by the inclusion $C_K \subseteq C_{K_1}$. It is therefore enough to show that the transfer

$$\mathrm{Ver} : G(K_1|K) \longrightarrow G(K_2|K_1)$$

is the trivial homomorphism. Since $K_1|K$ is the maximal unramified abelian extension of K in which the infinite places split completely, i.e., the maximal abelian subextension of $K_2|K$, we see that $G(K_2|K_1)$ is the commutator subgroup of $G(K_2|K)$. The proof of the principal ideal theorem is thus reduced to the following purely group-theoretic result. □

(7.6) Theorem. *Let G be a finitely generated group, G' its commutator subgroup, and G'' the commutator subgroup of G'. If $(G : G') < \infty$, then the transfer*

$$\mathrm{Ver} : G/G' \longrightarrow G'/G''$$

is the trivial homomorphism.

We give a proof of this theorem which is due to Ernst Witt [141]. In the group ring $\mathbb{Z}[G] = \{\sum_{\sigma\in G} n_\sigma \sigma \mid n_\sigma \in \mathbb{Z}\}$, we consider the **augmentation ideal** I_G, which is by definition the kernel of the ring homomorphism

$$\mathbb{Z}[G] \longrightarrow \mathbb{Z}, \quad \sum_\sigma n_\sigma \sigma \longmapsto \sum_\sigma n_\sigma .$$

For every subgroup H of G, we have $I_H \subseteq I_G$, and $\{\tau - 1 \mid \tau \in H, \tau \neq 1\}$ is a $\mathbb{Z}$-basis of I_H. We first establish the following lemma, which also has independent interest in that it gives an additive interpretation of the transfer.

(7.7) Lemma. *For every subgroup H of finite index in G, one has a commutative diagram*

$$\begin{array}{ccc} G/G' & \xrightarrow{\text{Ver}} & H/H' \\ {\scriptstyle\delta}\downarrow\cong & & {\scriptstyle\delta}\downarrow\cong \\ I_G/I_G^2 & \xrightarrow{S} & (I_H + I_G I_H)/I_G I_H, \end{array}$$

where the homomorphisms δ are induced by $\sigma \mapsto \delta\sigma = \sigma - 1$, and the homomorphism S is given by

$$S(x \bmod I_G^2) = x \sum_{\rho \in R} \rho \bmod I_G I_H,$$

for a system of representatives of the left cosets $R \ni 1$ of G/H.

Proof: We first show that the homomorphism

$$(*) \qquad H/H' \xrightarrow{\delta} (I_H + I_G I_H)/I_G I_H$$

induced by $\tau \mapsto \delta\tau = \tau - 1$ has an inverse. The elements $\rho\delta\tau$, $\tau \in H$, $\tau \neq 1$, $\rho \in R$, form a $\mathbb{Z}$-basis of $I_H + I_G I_H$. Indeed, it follows from

$$\rho\delta\tau = \delta\tau + \delta\rho\delta\tau$$

that they generate $I_H + I_G I_H$, and if

$$0 = \sum_{\rho,\tau} n_{\rho,\tau}\rho\delta\tau = \sum_{\rho,\tau} n_{\rho,\tau}(\rho\tau - \rho) = \sum_{\rho,\tau} n_{\rho,\tau}\rho\tau - \sum_{\rho}\Big(\sum_{\tau} n_{\rho,\tau}\Big)\rho,$$

then we conclude that $n_{\rho,\tau} = 0$ because the $\rho\tau, \rho$ are pairwise distinct. Mapping $\rho\delta\tau$ to $\tau \bmod H'$, we now have a surjective homomorphism

$$I_H + I_G I_H \longrightarrow H/H'.$$

It sends $\delta(\rho\tau')\delta\tau \in I_G I_H$ to $\tau'\tau\tau'^{-1}\tau^{-1} \equiv 1 \bmod H'$ because $\delta(\rho\tau')\delta\tau = \rho\delta(\tau'\tau) - \rho\delta\tau' - \delta\tau$. It thus induces a homomorphism which is inverse to $(*)$. In particular, if $H = G$, we obtain the isomorphism $G/G' \xrightarrow{\delta} I_G/I_G^2$.

The transfer is now obtained as

$$\text{Ver}(\sigma \bmod G') = \prod_{\rho \in R} \sigma_\rho \bmod H',$$

where $\sigma_\rho \in H$ is defined by $\sigma\rho = \rho'\sigma_\rho$, $\rho' \in R$. Ver thus induces the homomorphism

$$S : I_G/I_G^2 \longrightarrow (I_H + I_G I_H)/I_G I_H$$

given by $S(\delta\sigma \bmod I_G^2) = \sum_{\rho\in R} \delta\sigma_\rho \bmod I_G I_H$. From $\sigma\rho = \rho'\sigma_\rho$ follows the identity

$$\delta\rho + (\delta\sigma)\rho = \delta\sigma_\rho + \delta\rho' + \delta\rho'\delta\sigma_\rho .$$

Since ρ' runs through the set R if ρ does, we get as claimed

$$S(\delta\rho \bmod I_G^2) \equiv \sum_{\rho\in R} \delta\sigma_\rho \equiv \sum_{\rho\in R} (\delta\sigma)\rho \equiv \delta\sigma \sum_{\rho\in R} \rho \bmod I_G I_H . \qquad \square$$

Proof of theorem (7.6): Replacing G by G/G'', we may assume that $G'' = \{1\}$, i.e., that G' is abelian. Let $R \ni 1$ be a system of representatives of left cosets of G/G', and let $\sigma_1, \dots, \sigma_n$ be generators of G. Mapping $e_i = (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{Z}^n$ to σ_i, we get an exact sequence

$$0 \longrightarrow \mathbb{Z}^n \xrightarrow{f} \mathbb{Z}^n \longrightarrow G/G' \longrightarrow 1 ,$$

where f is given by an $n \times n$-matrix (m_{ik}) with $\det(m_{ik}) = (G : G')$. Consequently,

$$\prod_{i=1}^{n} \sigma_i^{m_{ik}} \tau_k = 1 \quad \text{with} \quad \tau_k \in G' .$$

The formulae $\delta(xy) = \delta x + \delta y + \delta x \delta y$, $\delta(x^{-1}) = -(\delta x)x^{-1}$ yield by iteration that

$$\delta\Big(\prod_{i=1}^{n} \sigma_i^{m_{ik}} \tau_k\Big) = \sum_{i=1}^{n} (\delta\sigma_i)\mu_{ik} = 0,$$

where $\mu_{ik} \equiv m_{ik} \bmod I_G$. In fact, the τ_k are products of commutators of the σ_i and σ_i^{-1}. We view (μ_{ik}) as a matrix over the commutative ring

$$\mathbb{Z}[G/G'] \cong \mathbb{Z}[G]/\mathbb{Z}[G]I_{G'},$$

which gives a meaning to the determinant $\mu = \det(\mu_{ik}) \in \mathbb{Z}[G/G']$. Let (λ_{kj}) be the adjoint matrix of (μ_{ik}). Then

$$(\delta\sigma_j)\mu = \sum_{i,k} (\delta\sigma_i)\mu_{ik}\lambda_{kj} \equiv 0 \bmod I_G \mathbb{Z}[G] I_{G'} ,$$

so that $(\delta\sigma)\mu \equiv 0 \bmod I_G\mathbb{Z}[G]I_{G'} = I_G I_{G'}$ for all σ. This yields

$$\mu \equiv \sum_{\rho\in R} \rho \bmod \mathbb{Z}[G]I_{G'} .$$

For if we put $\mu = \sum_{\rho\in R} n_\rho \overline{\rho}$, where $\overline{\rho} = \rho \bmod G'$, then for all $\overline{\sigma} \in G/G'$,

$$\overline{\sigma}\mu = \sum_{\rho} n_\rho \overline{\sigma}\,\overline{\rho} = \sum_{\rho} n_\rho \overline{\rho} .$$

This implies that all n_ρ are equal, hence $\mu \equiv m \sum_{\rho\in R} \rho \bmod \mathbb{Z}[G]I_{G'}$, and as

$$\mu \equiv \det(m_{ik}) \equiv (G : G') \equiv m(G : G') \bmod I_G,$$

we even have $m = 1$. Applying now lemma (7.7), we see that the transfer is the trivial homomorphism since

$$S(\delta\sigma \bmod I_G^2) \equiv \delta\sigma \sum_{\rho\in R} \rho \equiv (\delta\sigma)\mu \equiv 0 \bmod I_G I_{G'}. \qquad \square$$

A problem which is closely related to the principal ideal theorem and which was first put forward by *Philipp Furtwängler* is the **problem of the class field tower**. This is the question whether the class field tower

$$K = K_0 \subseteq K_1 \subseteq K_2 \subseteq K_3 \subseteq \dots,$$

where K_{i+1} is the Hilbert class field of K_i, stops after a finite number of steps. A positive answer would have the implication that the last field in the tower had class number 1 so that in it not only the ideals of K, but in fact all its ideals become principal. This perspective naturally generated the greatest interest. But the problem, after withstanding for a long time all attempts to solve it, was finally decided in the negative by the Russian mathematicians *E.S. Golod* and *I.R. Šafarevič* in 1964 (see [48], [24]).

Exercise 1. The decomposition law for the prime ideals $\mathfrak{p}$ which are *ramified* in an abelian extension $L|K$ can be formulated like this. Let $\mathfrak{f}$ be the conductor of $L|K$, $H^{\mathfrak{f}} \subseteq J_K^{\mathfrak{f}}$ the ideal group for L, and $H_{\mathfrak{p}}$ the smallest ideal group containing $H^{\mathfrak{f}}$ of conductor prime to $\mathfrak{p}$.

If $e = (H_{\mathfrak{p}} : H^{\mathfrak{f}})$ and $\mathfrak{p}^f$ is the smallest power of $\mathfrak{p}$ which belongs to $H_{\mathfrak{p}}$, then

$$\mathfrak{p} = (\mathfrak{P}_1 \cdots \mathfrak{P}_r)^e,$$

where the $\mathfrak{P}_i$ are of degree f over K, and $r = \frac{n}{ef}$, $n = [L : K]$.

Hint: The class field for $H_{\mathfrak{p}}$ is the inertia field above $\mathfrak{p}$.

The following exercises 2–6 concern a non-abelian example of *E. Artin*.

Exercise 2. The polynomial $f(X) = X^5 - X + 1$ is irreducible. The discriminant of a root α (i.e., the discriminant of $\mathbb{Z}[\alpha]$) is $d = 19 \cdot 151$.

Hint: The discriminant of a root of $X^5 + aX + b$ is $5^5b^4 + 2^8a^5$.

Exercise 3. Let $k = \mathbb{Q}(\alpha)$. Then $\mathbb{Z}[\alpha]$ is the ring $\mathcal{O}_k$ of integers of k.

Hint: The discriminant of $\mathbb{Z}[\alpha]$ equals the discriminant of $\mathcal{O}_k$ because on the one hand, both differ only by a square, and on the other hand, it is squarefree. The transition matrix from $1, \alpha, \dots, \alpha^{n-1}$ to an integral basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}_k$ is therefore invertible over $\mathbb{Z}$.

Exercise 4. The decomposition field $K\,|\,\mathbb{Q}$ of $f(X)$ has as Galois group the symmetric group $\mathfrak{S}_5$, i.e., it is of degree 120.

Exercise 5. K has class number 1.

Hint: Show, using chap. I, § 6, exercise 3, that every ideal class of K contains an ideal $\mathfrak{a}$ with $\mathfrak{N}(\mathfrak{a}) < 4$. If $\mathfrak{N}(\mathfrak{a}) \neq 1$, then $\mathfrak{a}$ has to be a prime ideal $\mathfrak{p}$ such that $\mathfrak{N}(\mathfrak{p}) = 2$ or 3. Hence $\mathcal{O}_k/\mathfrak{p} = \mathbb{Z}/2\mathbb{Z}$ or $= \mathbb{Z}/3\mathbb{Z}$, so f has a root mod 2 or 3, which is not the case.

Exercise 6. Show that $K\,|\,\mathbb{Q}(\sqrt{19\cdot 151})$ is a (non-abelian!) unramified extension.

Exercise 7. For every Galois extension $L|K$ of finite algebraic number fields, there exist infinitely many finite extensions K' such that $L \cap K' = K$, and such that $LK'|K'$ is unramified.

Hint: Let S be the set of places ramified in $L|K$, and let $L_\mathfrak{p} = K_\mathfrak{p}(\alpha_\mathfrak{p})$. By the approximation theorem, choose an algebraic number α which, for every $\mathfrak{p} \in S$, is close to $\alpha_\mathfrak{p}$ when embedded into $\overline{K}_\mathfrak{p}$. Then $K_\mathfrak{p}(\alpha_\mathfrak{p}) \subseteq K_\mathfrak{p}(\alpha)$ by Krasner's lemma, chap. II, § 6, exercise 2. Put $K' = K(\alpha)$ and show that $LK'|K'$ is unramified. To show that α can be chosen such that $L \cap K' = K$ use (3.7), and the fact that $G(L|K)$ is generated by elements of prime power order.

§ 8. The Reciprocity Law of the Power Residues

In class field theory Gauss's reciprocity law meets its most general and definite formulation. Let n be a positive integer ≥ 2 and K a number field containing the group μ_n of n-th roots of unity. In chap. V, § 3, we introduced, for every place $\mathfrak{p}$ of K, the n-th Hilbert symbol

$$\left(\frac{\ ,\ }{\mathfrak{p}}\right) : K_\mathfrak{p}^* \times K_\mathfrak{p}^* \longrightarrow \mu_n.$$

It is given via the norm residue symbol by

$$\big(a, K_\mathfrak{p}(\sqrt[n]{b})|K_\mathfrak{p}\big)\sqrt[n]{b} = \left(\frac{a,b}{\mathfrak{p}}\right)\sqrt[n]{b}.$$

These symbols all fit together in the following *product formula*.

(8.1) Theorem. *For $a, b \in K^*$ one has*

$$\prod_\mathfrak{p}\left(\frac{a,b}{\mathfrak{p}}\right) = 1.$$

Proof: From (5.7), we find

$$\Big[\prod_{\mathfrak{p}}\Big(\frac{a,b}{\mathfrak{p}}\Big)\Big]\sqrt[n]{b} = \Big[\prod_{\mathfrak{p}}(a, K_{\mathfrak{p}}(\sqrt[n]{b})|K_{\mathfrak{p}})\Big]\sqrt[n]{b} = \big(a, K(\sqrt[n]{b})|K\big)\sqrt[n]{b} = \sqrt[n]{b},$$

and hence the theorem. □

In chap. V, § 3, we defined the n-th power residue symbol in terms of the Hilbert symbol:

$$\Big(\frac{a}{\mathfrak{p}}\Big) = \Big(\frac{\pi, a}{\mathfrak{p}}\Big),$$

where $\mathfrak{p}$ is a prime ideal of K not dividing n, $a \in U_{\mathfrak{p}}$, and π is a prime element of $K_{\mathfrak{p}}$. We have seen that this definition does not depend on the choice of the prime element π and that one has

$$\Big(\frac{a}{\mathfrak{p}}\Big) = 1 \iff a \equiv \alpha^n \bmod \mathfrak{p},$$

and more generally

$$\Big(\frac{a}{\mathfrak{p}}\Big) \equiv a^{(q-1)/n} \bmod \mathfrak{p}, \quad q = \mathfrak{N}(\mathfrak{p}).$$

(8.2) Definition. *For every ideal* $\mathfrak{b} = \prod_{\mathfrak{p}\nmid n} \mathfrak{p}^{\nu_{\mathfrak{p}}}$ *prime to* n, *and every number* a *prime to* $\mathfrak{b}$, *we define the* ***n*-th power residue symbol** *by*

$$\Big(\frac{a}{\mathfrak{b}}\Big) = \prod_{\mathfrak{p}\nmid n}\Big(\frac{a}{\mathfrak{p}}\Big)^{\nu_{\mathfrak{p}}}.$$

Here $\Big(\frac{a}{\mathfrak{p}}\Big)^{\nu_{\mathfrak{p}}} = 1$ *when* $\nu_{\mathfrak{p}} = 0$.

The power residue symbol $\big(\frac{a}{\mathfrak{b}}\big)$ is obviously multiplicative in both arguments. If $\mathfrak{b}$ is a principal ideal (b), we write for short $\big(\frac{a}{\mathfrak{b}}\big) = \big(\frac{a}{b}\big)$. We now prove the **general reciprocity law for the *n*-th power residues.**

(8.3) Theorem. *If* $a, b \in K^*$ *are prime to each other and to* n, *then*

$$\Big(\frac{a}{b}\Big)\Big(\frac{b}{a}\Big)^{-1} = \prod_{\mathfrak{p}|n\infty}\Big(\frac{a,b}{\mathfrak{p}}\Big).$$

Proof: If $\mathfrak{p}$ is prime to $bn\infty$, then we have

$$\left(\frac{b}{\mathfrak{p}}\right)^{v_\mathfrak{p}(a)} = \left(\frac{\pi, b}{\mathfrak{p}}\right)^{v_\mathfrak{p}(a)} = \left(\frac{a, b}{\mathfrak{p}}\right),$$

where π is a prime element of $K_\mathfrak{p}$. For if we put $a = u\pi^{v_\mathfrak{p}(a)}$, then $(\frac{u,b}{\mathfrak{p}}) = 1$ because $u, b \in U_\mathfrak{p}$. For the same reason, we find

$$\left(\frac{a,b}{\mathfrak{p}}\right) = 1 \quad \text{for } \mathfrak{p} \text{ prime to } abn\infty.$$

(8.1) then gives

$$\begin{aligned}\left(\frac{a}{b}\right)\left(\frac{b}{a}\right)^{-1} &= \prod_{\mathfrak{p}|(b)}\left(\frac{a}{\mathfrak{p}}\right)^{v_\mathfrak{p}(b)}\prod_{\mathfrak{p}|(a)}\left(\frac{b}{\mathfrak{p}}\right)^{-v_\mathfrak{p}(a)} = \prod_{\mathfrak{p}|(b)}\left(\frac{b,a}{\mathfrak{p}}\right)\prod_{\mathfrak{p}|(a)}\left(\frac{a,b}{\mathfrak{p}}\right)^{-1}\\ &= \prod_{\mathfrak{p}|(ab)}\left(\frac{b,a}{\mathfrak{p}}\right) = \prod_{\mathfrak{p}\nmid n\infty}\left(\frac{b,a}{\mathfrak{p}}\right) = \prod_{\mathfrak{p}|n\infty}\left(\frac{a,b}{\mathfrak{p}}\right).\end{aligned}$$

Here $\mathfrak{p}|(b)$ means that $\mathfrak{p}$ occurs in the prime decomposition of (b). □

Gauss's reciprocity law, for which we gave an elementary proof using the theory of Gauss sums in chap. I, (8.6), in the case of two odd prime numbers p, l, is contained in the general reciprocity law (8.3) as a special case. For if we substitute, in the case $K = \mathbb{Q}$, $n = 2$, into formula (8.3) the explicit description (chap. V, (3.6)) of the Hilbert symbol $\left(\frac{a,b}{p}\right)$ for $p = 2$ and $p = \infty$, we obtain the following theorem, which is more general than chap. I, (8.6).

(8.4) Gauss's Reciprocity Law. *Let $K = \mathbb{Q}$, $n = 2$, and let a and b be odd, relatively prime integers. Then one has*

$$\left(\frac{a}{b}\right)\left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}(-1)^{\frac{\operatorname{sgn} a-1}{2}\frac{\operatorname{sgn} b-1}{2}},$$

and for positive odd integers b, we have the two "supplementary theorems"

$$\left(\frac{-1}{b}\right) = (-1)^{\frac{b-1}{2}}, \quad \left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}}.$$

For the last equation we need again the product formula:

$$\left(\frac{2}{b}\right) = \prod_{p\neq 2,\infty}\left(\frac{p,2}{p}\right)^{v_p(b)} = \prod_{p\neq 2,\infty}\left(\frac{b,2}{p}\right) = \left(\frac{2,b}{2}\right)\left(\frac{2,b}{\infty}\right) = (-1)^{\frac{b^2-1}{2}}.$$

The symbol $\left(\frac{a}{b}\right)$ is called the **Jacobi symbol**, or also the **quadratic residue symbol** (although, for b not a prime number, the condition that the symbol $\left(\frac{a}{b}\right) = 1$ is no longer equivalent to the condition that a is a quadratic residue modulo b).

In the above formulation, the reciprocity law allows us to compute simply by iteration the quadratic residue symbol $\left(\frac{a}{b}\right)$, as is shown in the following example:

$$\left(\frac{40077}{65537}\right) = \left(\frac{65537}{40077}\right) = \left(\frac{25460}{40077}\right) = \left(\frac{2^2}{40077}\right)\left(\frac{6365}{40077}\right) = \left(\frac{40077}{6365}\right) =$$
$$\left(\frac{1887}{6365}\right) = \left(\frac{6365}{1887}\right) = \left(\frac{704}{1887}\right) = \left(\frac{4^3}{1887}\right)\left(\frac{11}{1887}\right) = -\left(\frac{1887}{11}\right) =$$
$$-\left(\frac{6}{11}\right) = -\left(\frac{2}{11}\right)\left(\frac{3}{11}\right) = \left(\frac{3}{11}\right) = -\left(\frac{11}{3}\right) = -\left(\frac{2}{3}\right) = 1.$$

Class field theory originated from Gauss's reciprocity law. The quest for a similar law for the n-th power residues dominated number theory for a long time, and the all-embracing answer was finally found in Artin's reciprocity law. The above reciprocity law (8.3) of the power residues now appears as a simple and special consequence of Artin's reciprocity law. But to really settle the original problem, class field theory was still lacking the explicit computation of the Hilbert symbols $\left(\frac{a,b}{\mathfrak{p}}\right)$ for $\mathfrak{p} | n\infty$. This was finally completed in the 1960s by the mathematician *Helmut Brückner*, see chap. V, (3.7).